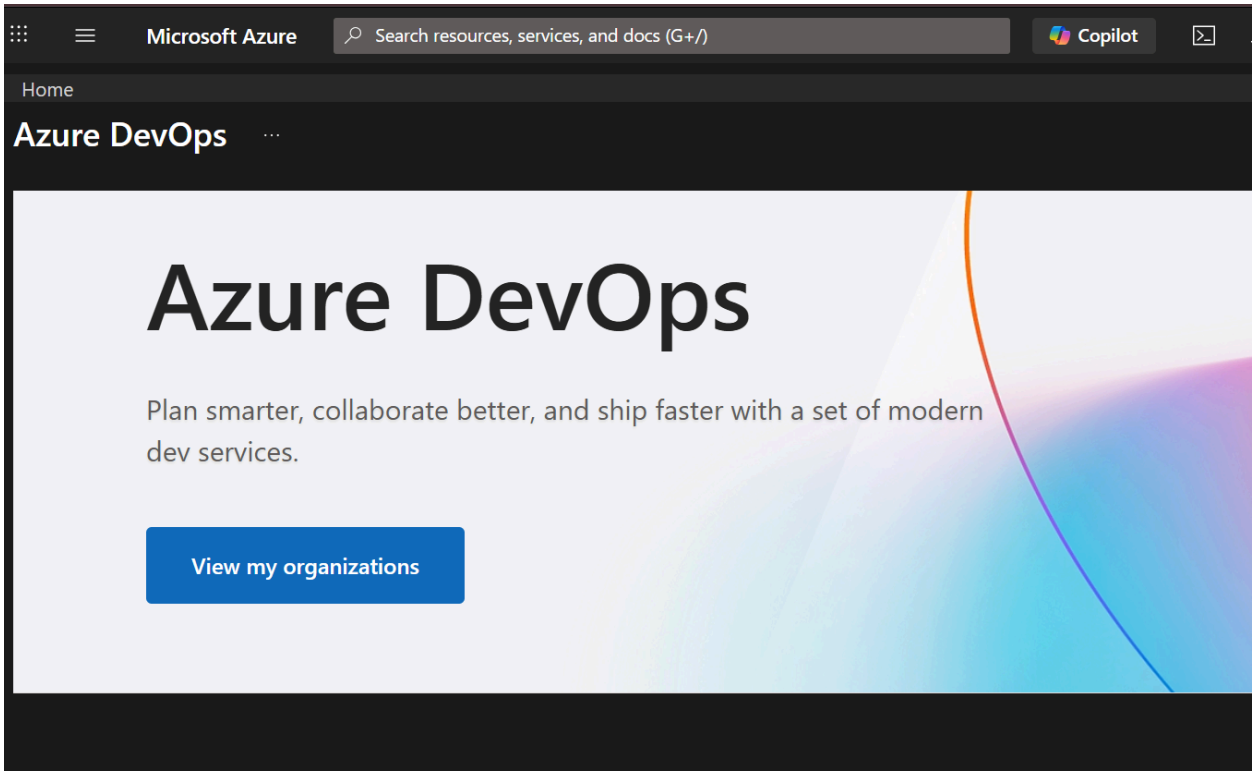
Azure cloud / portal

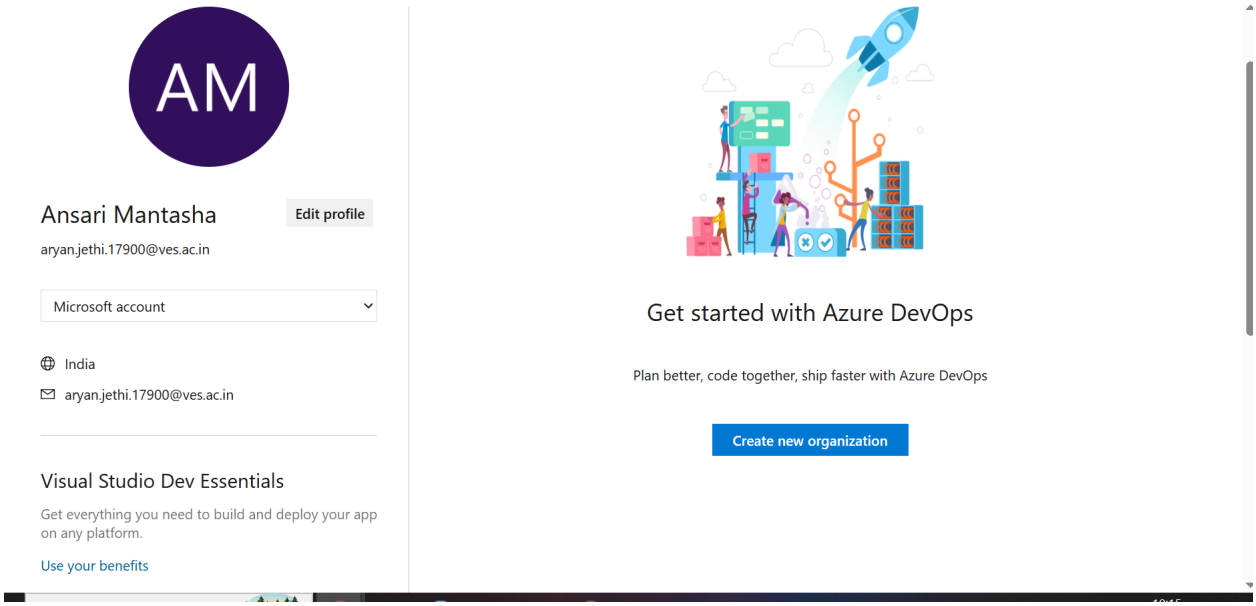
Azure devops
It s a platform that is integrated with tools like azure repo
>>repo
>> AZ pipeline
>>AZ artifacts
>>access management
>>templates
This project we are going to refer
https://github.com/jaiswaladi246/Boardgame/tree/main

Search for azure devops organizations



\vew my organizations

Enter you basic details



Create new org

Almost done...

Name your Azure DevOps organization *

dev.azure.com/ azure0pipeline

We'll host your projects in

India

Select an Azure subscription for billing

Azure for Students (3dd77353-9376-4da6-b87d...

You can also create a new Azure subscription

Enter the characters you see

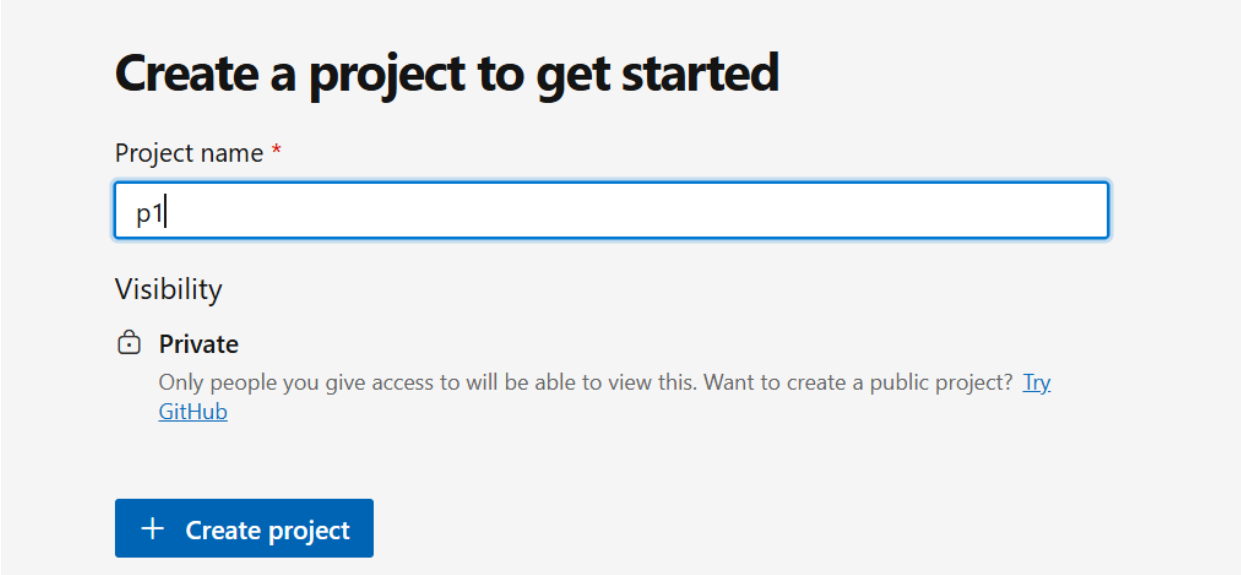
New Audio

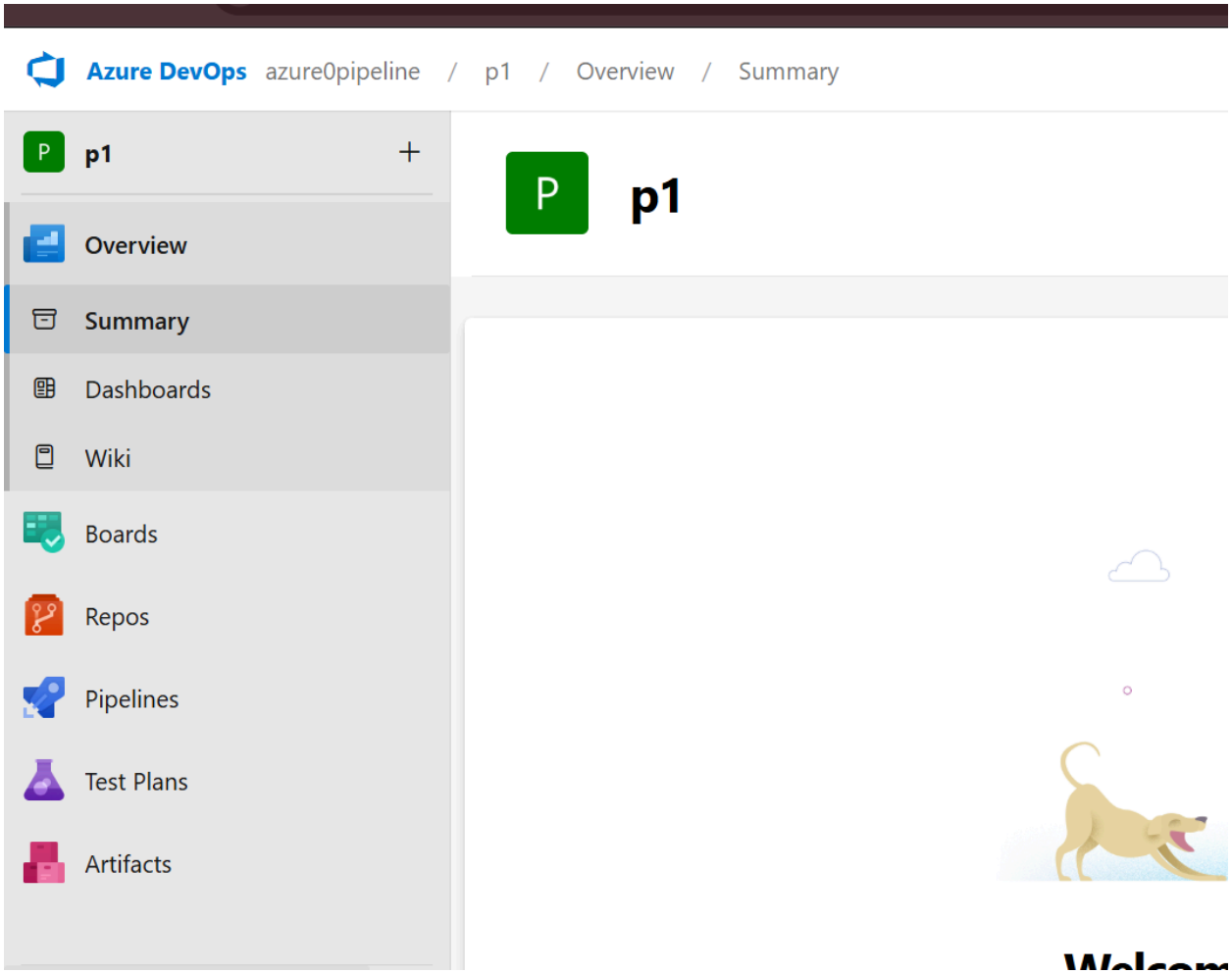
H3Vk

Continue

It will take few minutes and your organizatino will be created

In a single organaization you can create multiple projects





Now lets create a pipeline

📁 Overview

Quick summary of your project — shows basic info, recent activity, and health at a glance.

📊 Summary

A dashboard-style view of project status, pipelines, repos, and progress in one place.

📈 Dashboards

Customizable boards where you can pin charts, builds, and metrics to monitor your project.

📚 Wiki

Documentation area — write and store project notes, guides, and instructions.

📝 Boards

Used for **task management** — create work items, track bugs, and manage sprint progress (Agile/Kanban).>>ticketing tools

📝 **Work Items**: All your tasks, bugs, and features in one list — create and track work here.

📋 **Boards** : Kanban-style board to **move tasks** (To Do → In Progress → Done).

📚 **Backlogs** : A prioritized list of all work that needs to be done in the future.

🏃 **Sprints** : Short work cycles (like 1–2 weeks) where the team completes selected tasks.

🔍 **Queries** : Search and filter work items (e.g., “show all open bugs”).

📅 **Delivery Plans** : High-level timeline view of multiple teams’ work and progress.

📊 **Analytics Views** : Reports and charts to see progress, trends, and team performance

💻 Repos

Your **source code storage** (Git) — manage code, branches, and pull requests.

🚀 Pipelines

CI/CD automation — build, test, and deploy your application automatically.

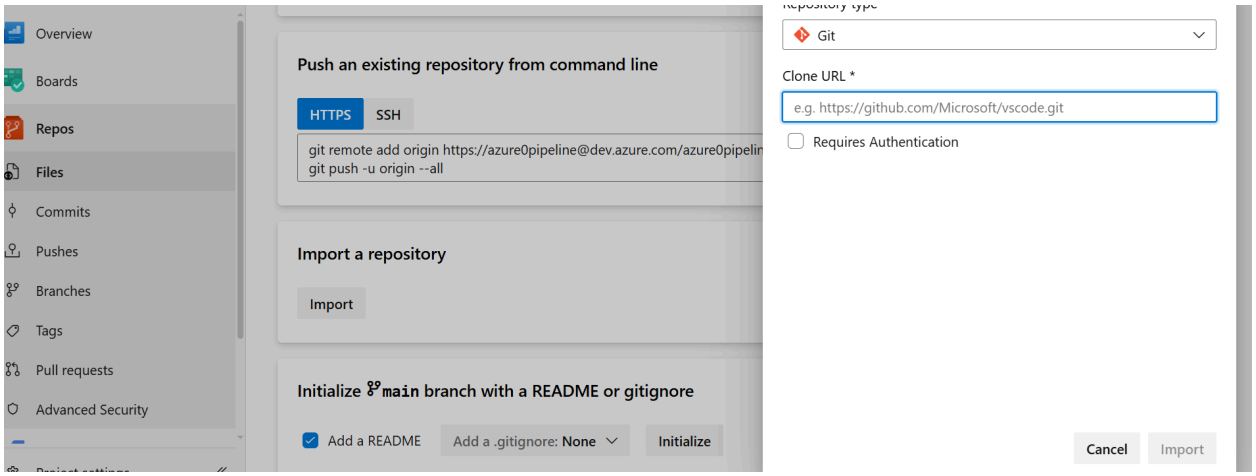
🧪 Test Plans

Manage manual and automated testing — create test cases and track results.

📦 Artifacts

Store and share build outputs like packages, libraries, and dependencies.

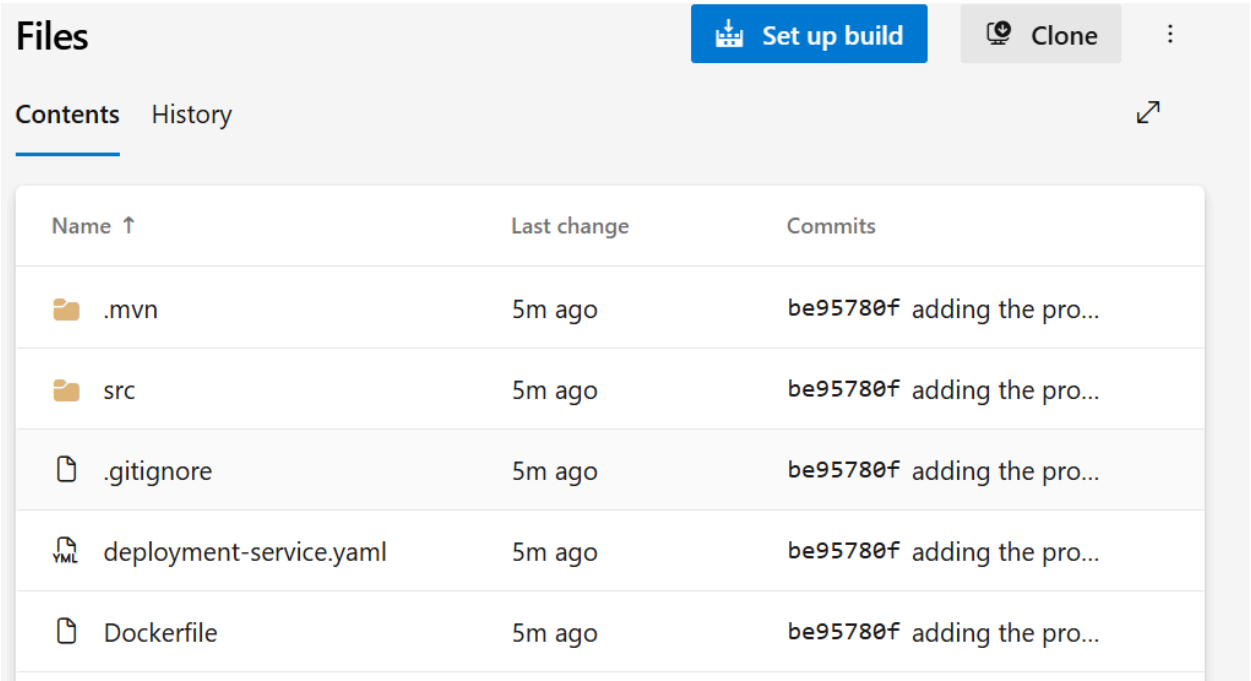
Now lets clone the repo inside the azure
Goto repo>>file >> import repo >> choose git



https://github.com/mantu0tech/Azure_Devops_CICD_Pipeline.git

And click on import it will clone your repo

And its clone your repo perfectly



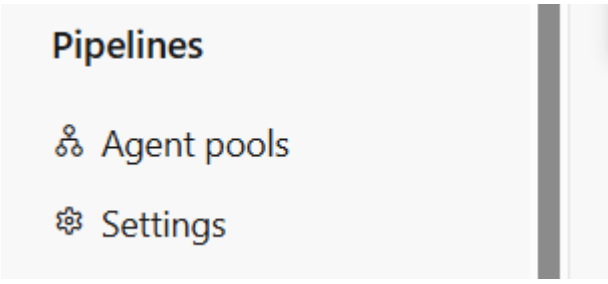
Make your own repo and push the code and make your own image locally and push it to docker hub
In azure we have 2 types of pipeline

Classic pipeline yaml pipeline

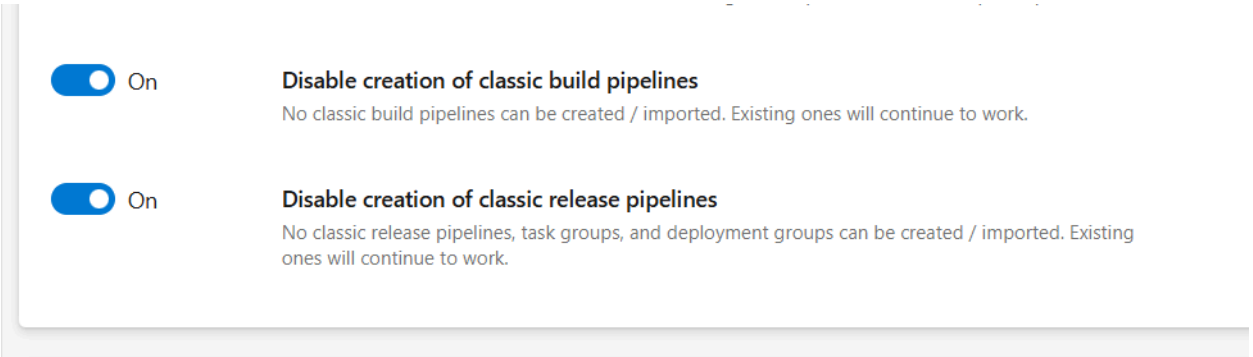
Yaml is the easiest ways to to do it

Pipeline create pipeline>>

We are going to classic
To get the both optics

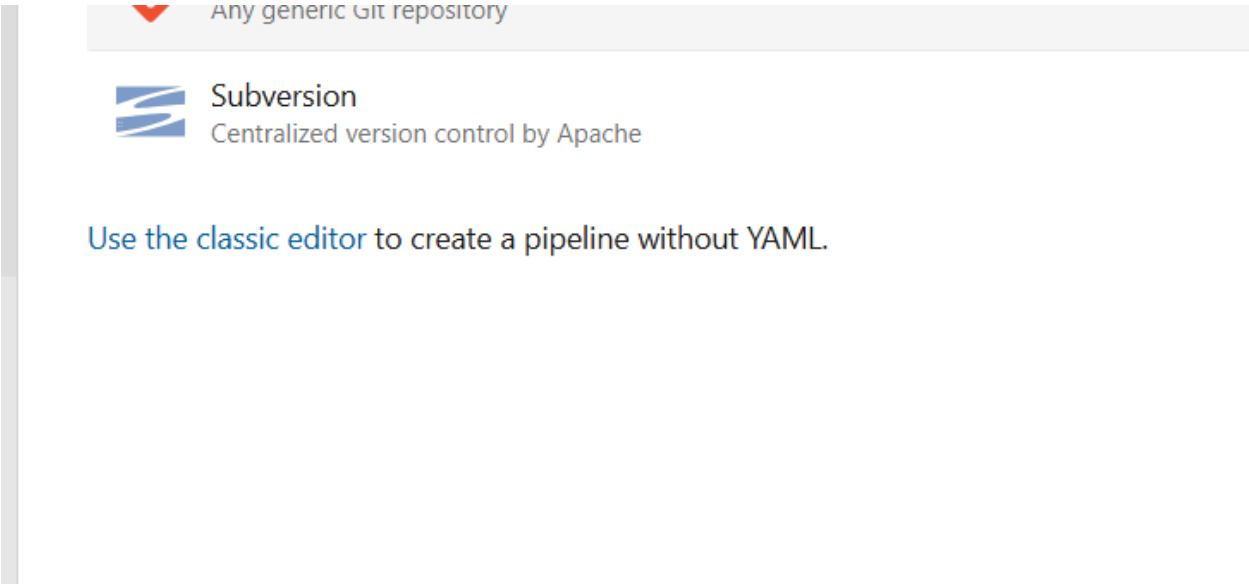
Go to org> org setting >> >> 

Disable these 2 optis o

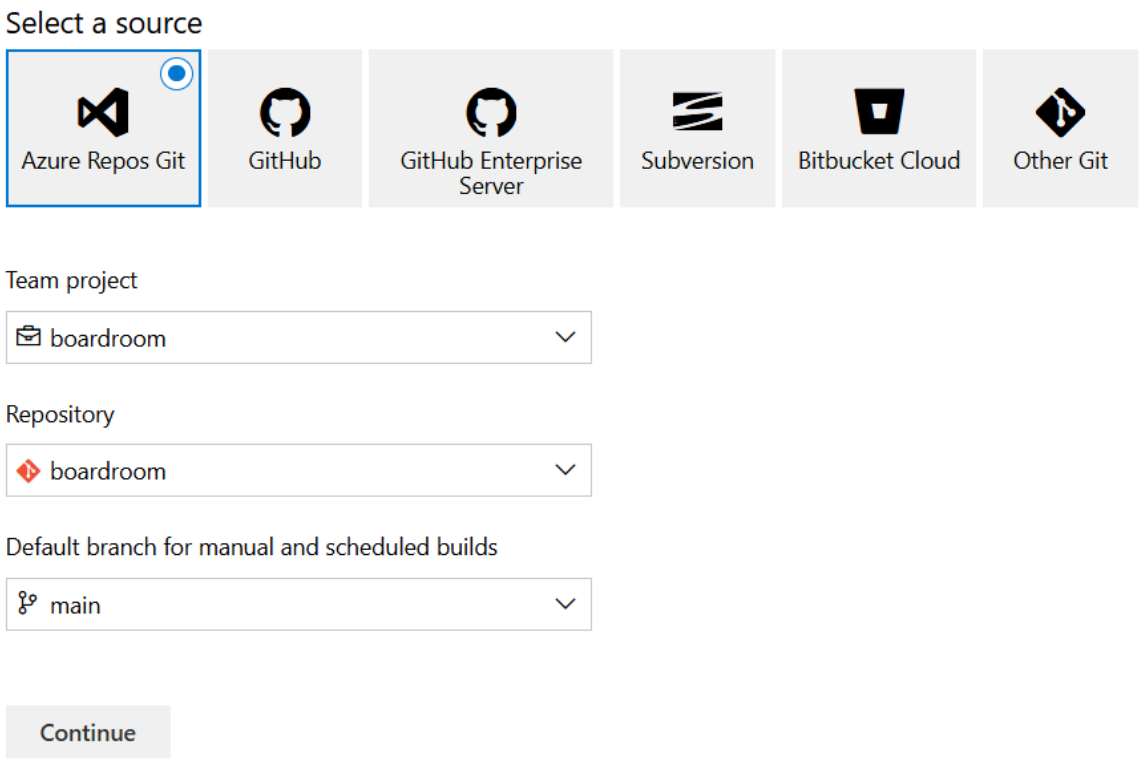


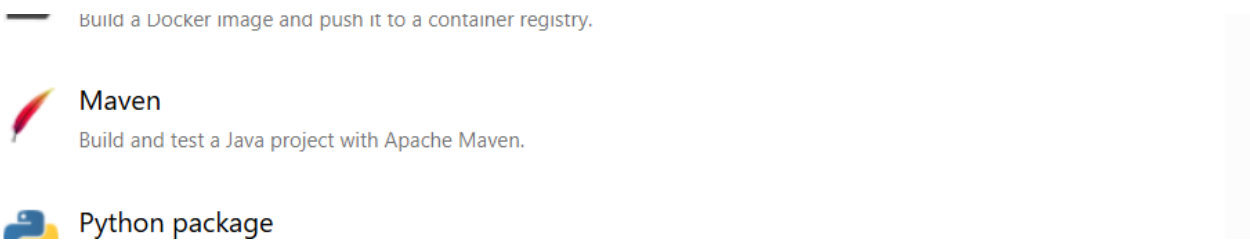
]now will have classic options
Each project is isolated from each others

Create pipeline



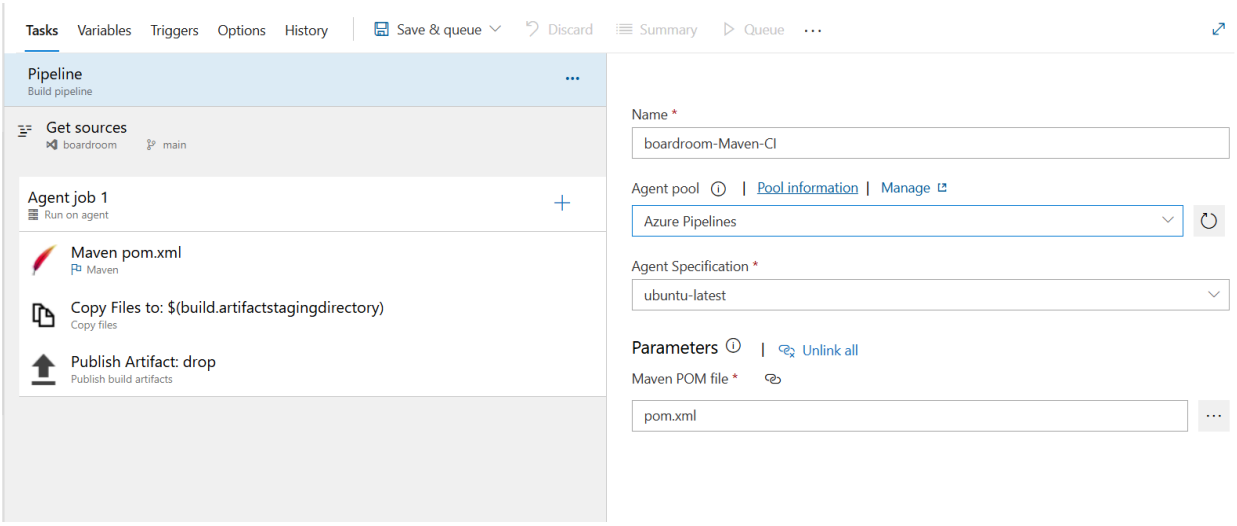
Click on link andcreate

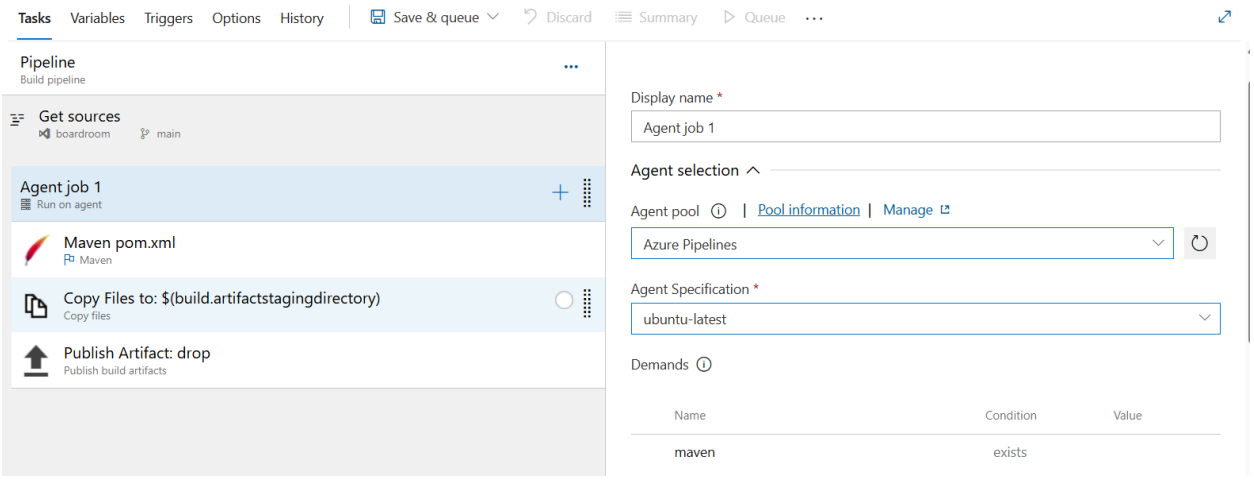


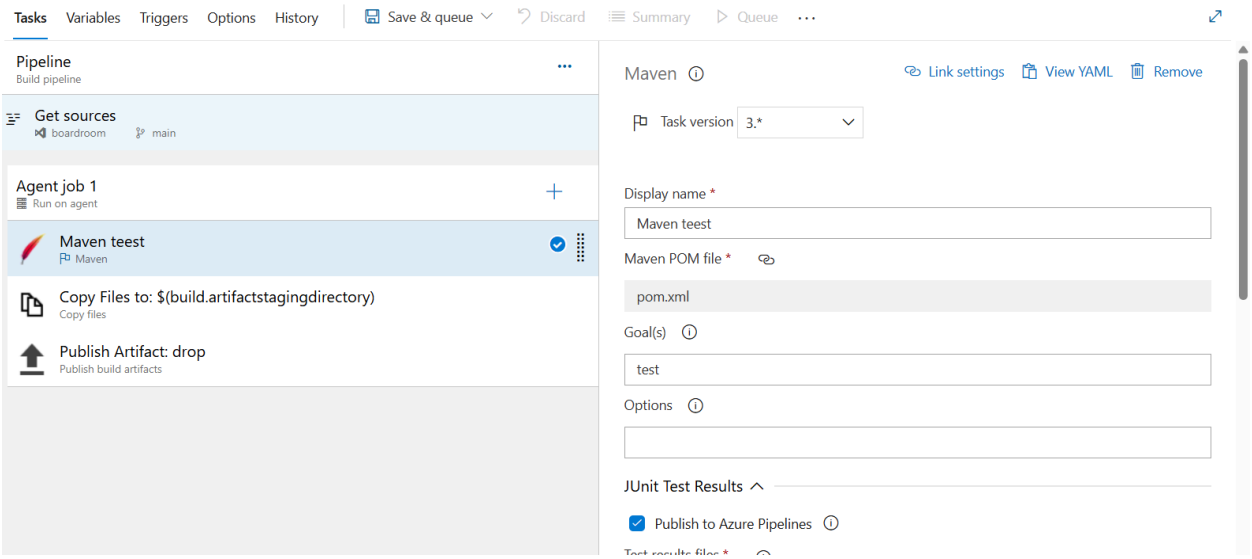


Go with maven as our project is based on maven

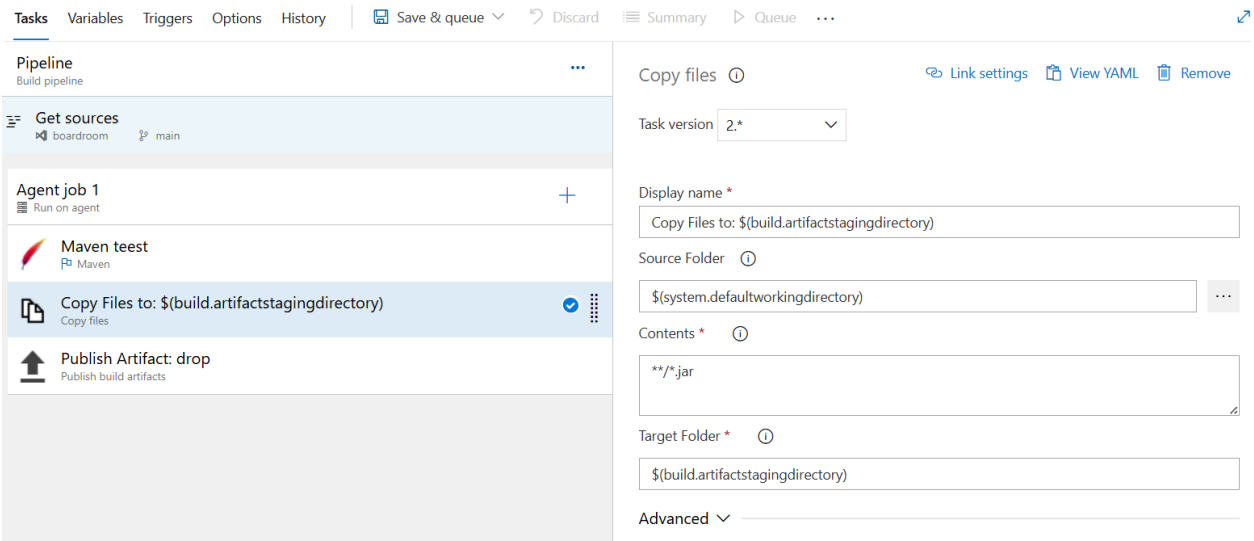
We have agent pool that means agent or slave >> azure pipeline > so here we can have



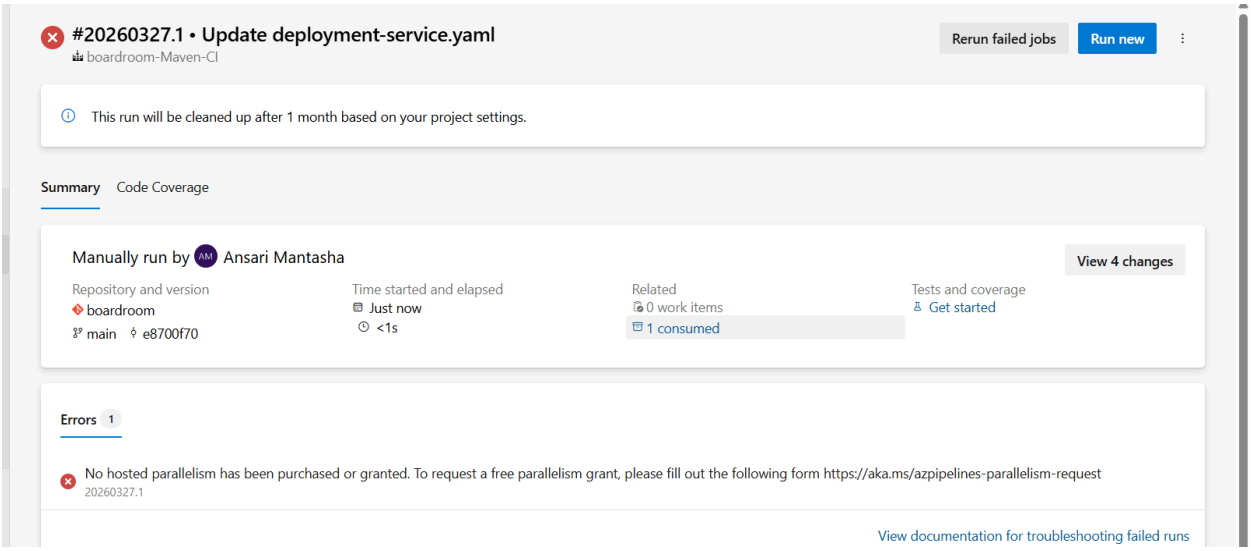




It will do everything by it self so we just need t define it



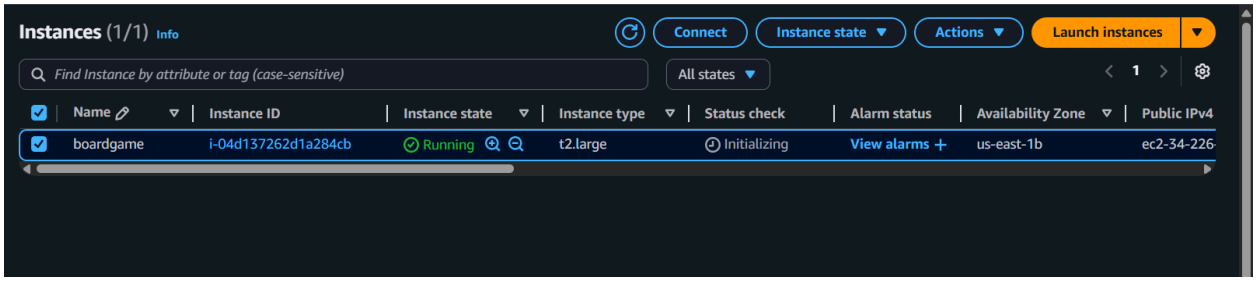
Classic option will do al the work by self
If
Now save and queue
And save and run



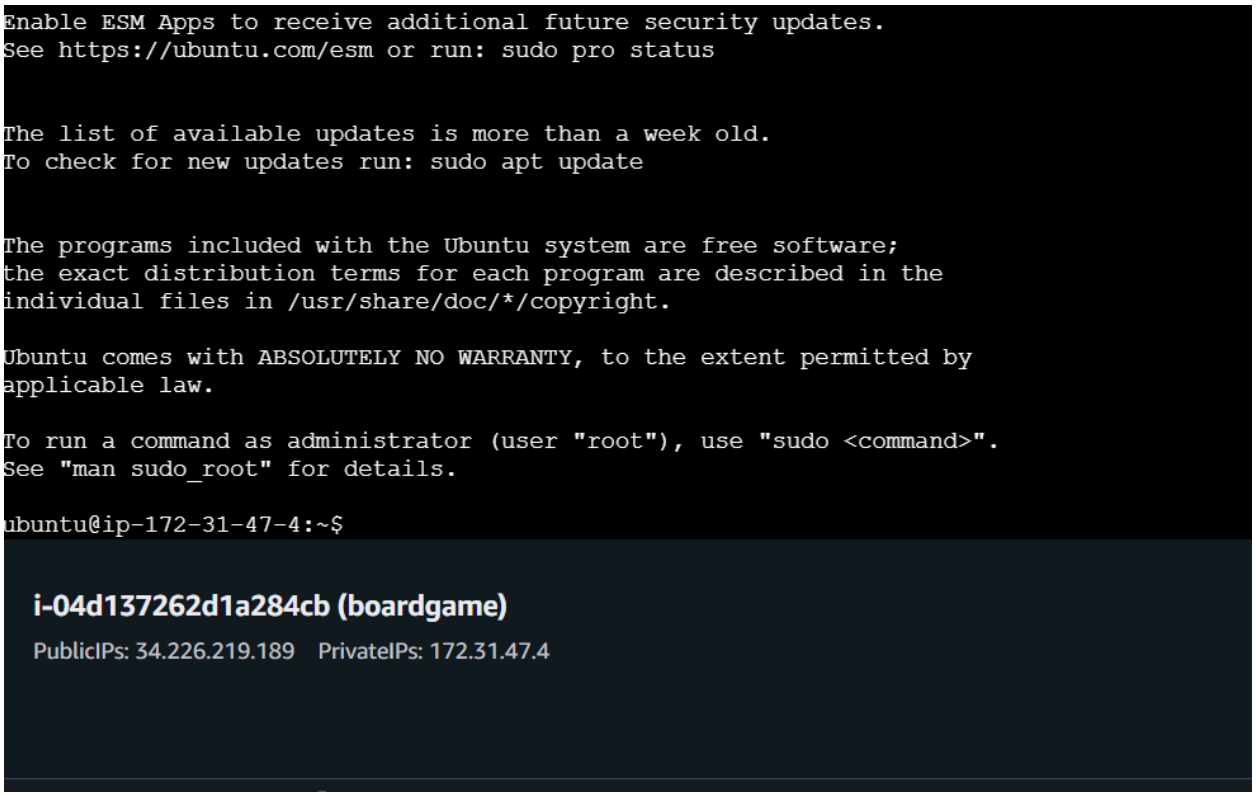
You can see theerro

We need the permission to run on azure agents
So will create or own agent and run on it

So lets create a vm
We are going to launc aws ec2 instance
\we are using the ubuntu >> t2 large
Storage is 25
Add port like 25,3000-10000
80,443,22,6443,465,27017,30000-32767 add all in sg



Now lets connect is



Now lets install everything

How to connect your vm to azure
Go to org >> setting >> agent pool

Click on add pools

And if you want for your particular project then g with the project setting and add

Add agent pool

Agent pools are shared across an organization.

Pool type:

- ○ Managed DevOps Pool

  Reduce the effort spent in maintaining custom agents by creating a Microsoft managed pool of scalable agents. Learn more.

- ◉ Self-hosted

  Create a pool of custom agents hosted on your own infrastructure for maximum control and flexibility. Learn more.

- ○ Azure virtual machine scale set

  Create a pool of custom agents based on an Azure Virtual machine scale set hosted in your own Azure subscription. View configuration instructions.

Name:

mantasha

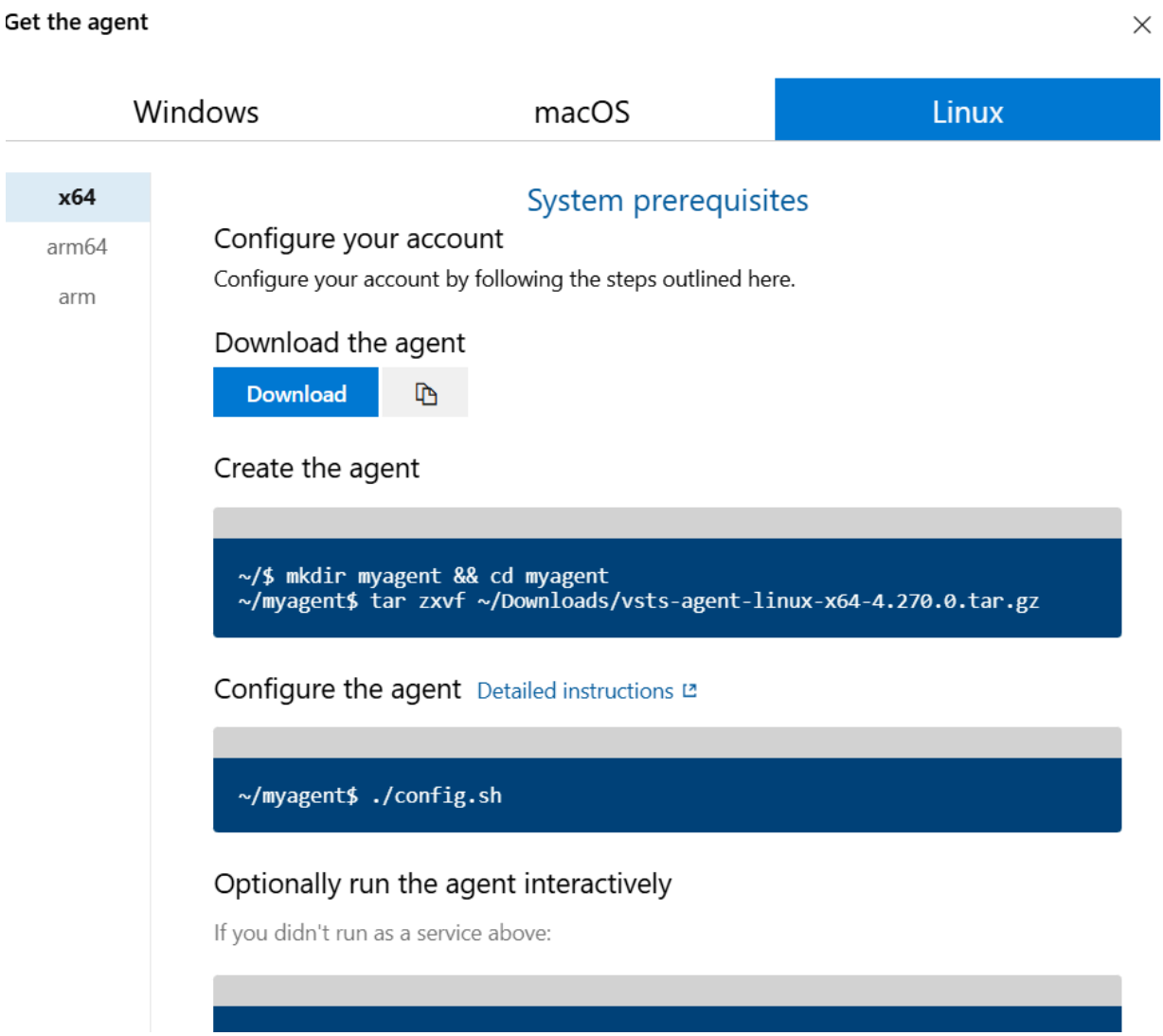
Description (optional):

Markdown supported.

Pipeline permissions:

☑ Auto-provision this agent pool in all projects

Create it
Go inside it and click on add >>new agents



Now arun all these in your server

Now run these thing now we are goin to connect

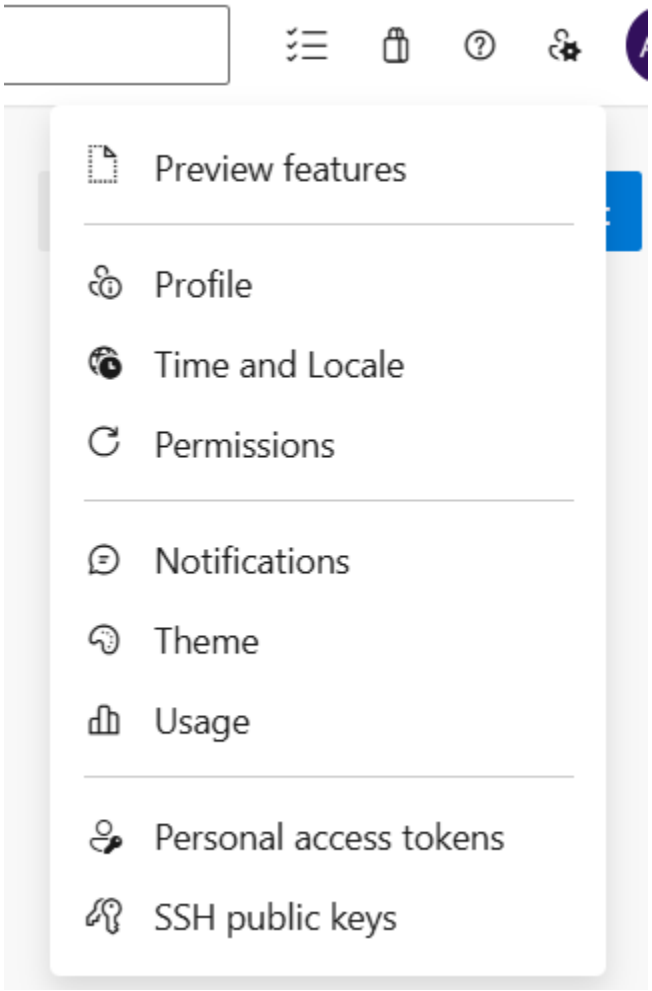
```
  1  sudo apt update
  2  wget
https://download.agent.dev.azure.com/agent/4.270.0/vsts-agent-linux-x64-4.270.0.tar.gz
  3  mkdir myagent && cd myagent
  6  cd ..
  7  ls
  8  mv vsts-agent-linux-x64-4.270.0.tar.gz myagent/
  9  cd myagent/
 10  ls
 11  tar -xvf vsts-agent-linux-x64-4.270.0.tar.gz
 12  history
```

ubuntu@ip-172-31-47-4:~/myagent$ ^C
ubuntu@ip-172-31-47-4:~/myagent$

./config.sh

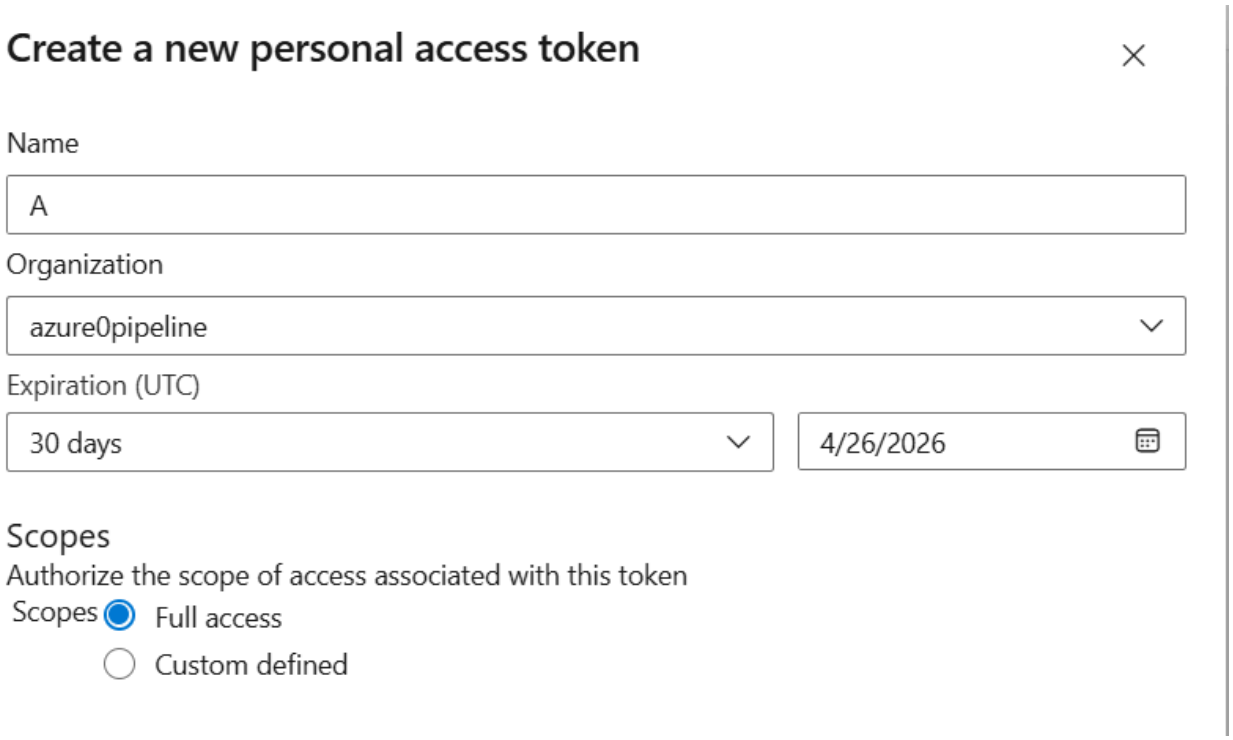
Type Y
Url >> https://dev.azure.com/azure0pipeline till your org give link

We are going to use PAT to connect



Go t oPAT
New token

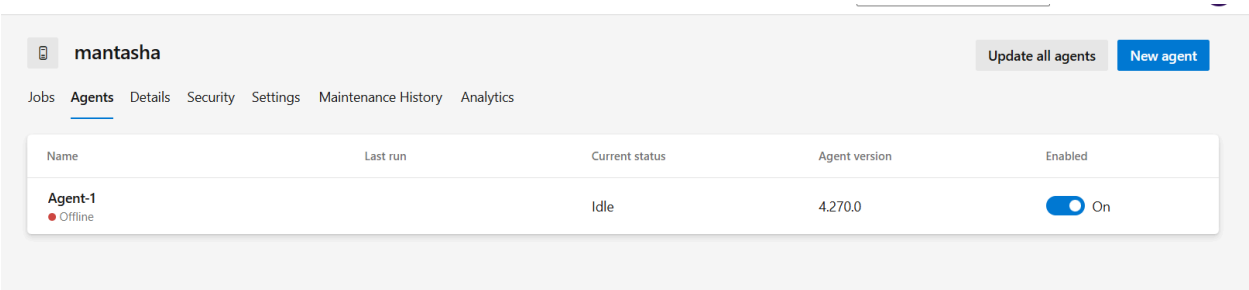
8jCN3YFagYkR9dzx64giOz6h71FkFSzDXfvTDtSc0ykaOtQXoPUsJQQJ99CCACAAAAAAAAAAAAASA
ZDO1bQcnonow



Now enter your pool name
And its mantasha

```
 Git repositories.

A copy of the Team Explorer Everywhere license agreement can be found at:
  /home/ubuntu/myagent/license.html

Enter (Y/N) Accept the Team Explorer Everywhere license agreement now? (press enter for N) > Y

>> Connect:

Enter server URL > https://dev.azure.com/azure0pipeline
Enter authentication type (press enter for PAT) > 8jCN3YFagYkR9dzx64giOz6h71FkFSzDXfvTDtSc0ykaOtQXoPUsJQQJ99CCACAAAAAAAAAAAAASAZDO1bQc
Enter a valid value for authentication type.
Enter authentication type (press enter for PAT) >
Enter personal access token > ********************************************************************************
Connecting to server ...

>> Register Agent:

Enter agent pool (press enter for default) > mantasha
Enter agent name (press enter for ip-172-31-47-4) > Agent-1
Scanning for tool capabilities.
Connecting to the server.
Successfully added the agent
Testing agent connection.
Enter work folder (press enter for _work) >
2026-03-27 08:41:12Z: Settings Saved.
ubuntu@ip-172-31-47-4:~/myagent$
```

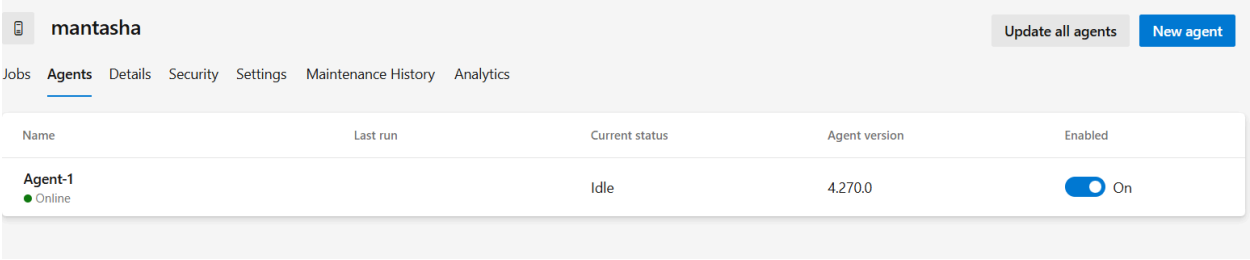
Now go inside your agent and your ca see it connected



To start the agent the these script

```
 sudo apt update
   2  wget
https://download.agent.dev.azure.com/agent/4.270.0/vsts-agent-linux-x64-4.270.0.tar.gz
   3  mkdir myagent && cd myagent
   5  ls
   6  cd ..
   7  ls
   8  mv vsts-agent-linux-x64-4.270.0.tar.gz myagent/
   9  cd myagent/
  10  ls
  11  tar -xvf vsts-agent-linux-x64-4.270.0.tar.gz
```

./[run.sh](run.sh)



Now stop these and install java maven docker >>
Sonar qube >> it wil check the code quality

https://hub.docker.com/r/mc1arke/sonarqube-with-community-branch-plugin
We are going to use these

Now we are goging to install triv
Now install it

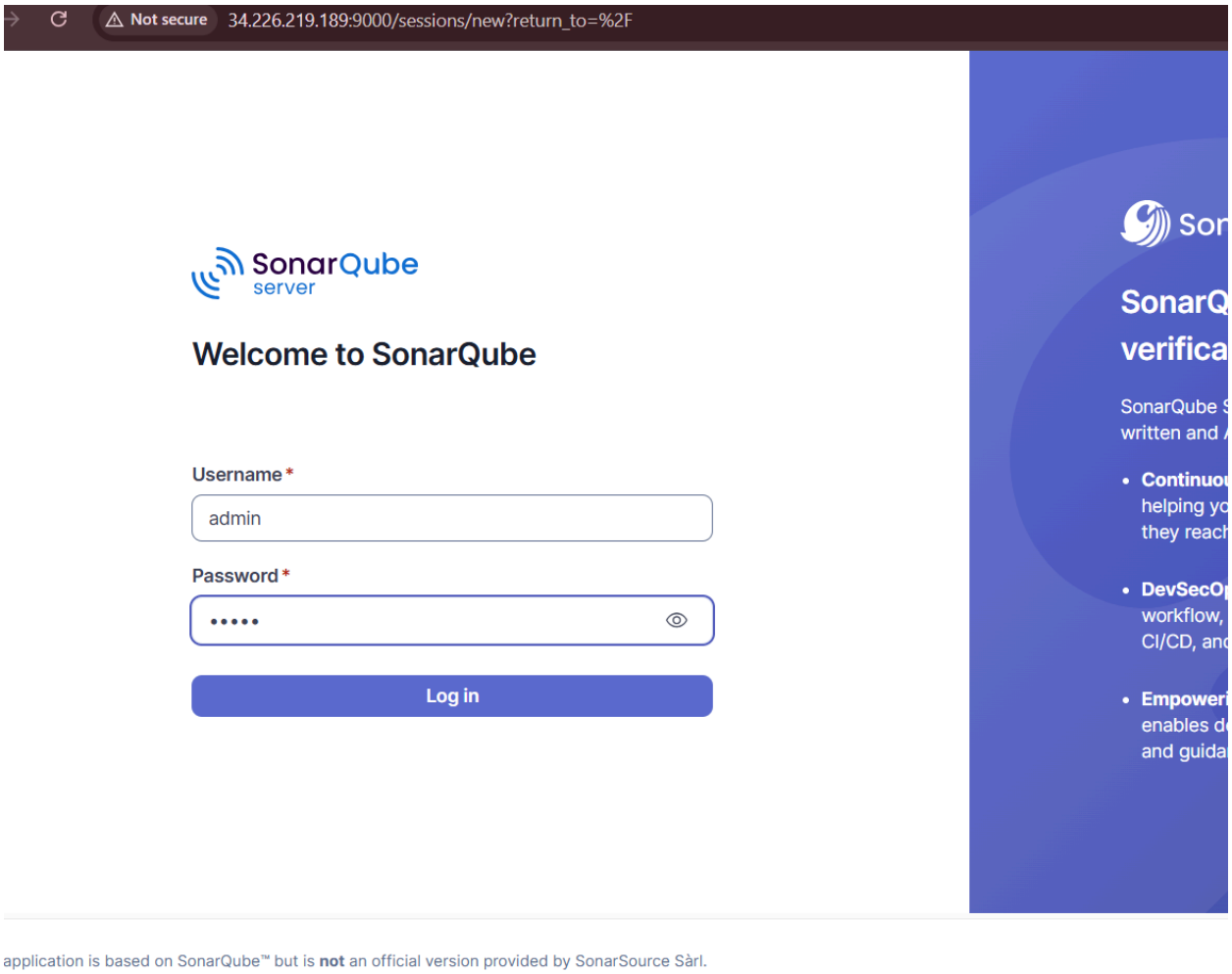
And install docker
 ./run.sh

```
sudo apt install openjdk-17-jre-headless -y
sudo apt install maven -y
java --version
maven --version
maven version
maven
mvn --version
trivy
sudo snap install trivy
trivy --version
sudo apt install docker.io -y
sudo usermod -aG docker ubuntu
```

Now run these

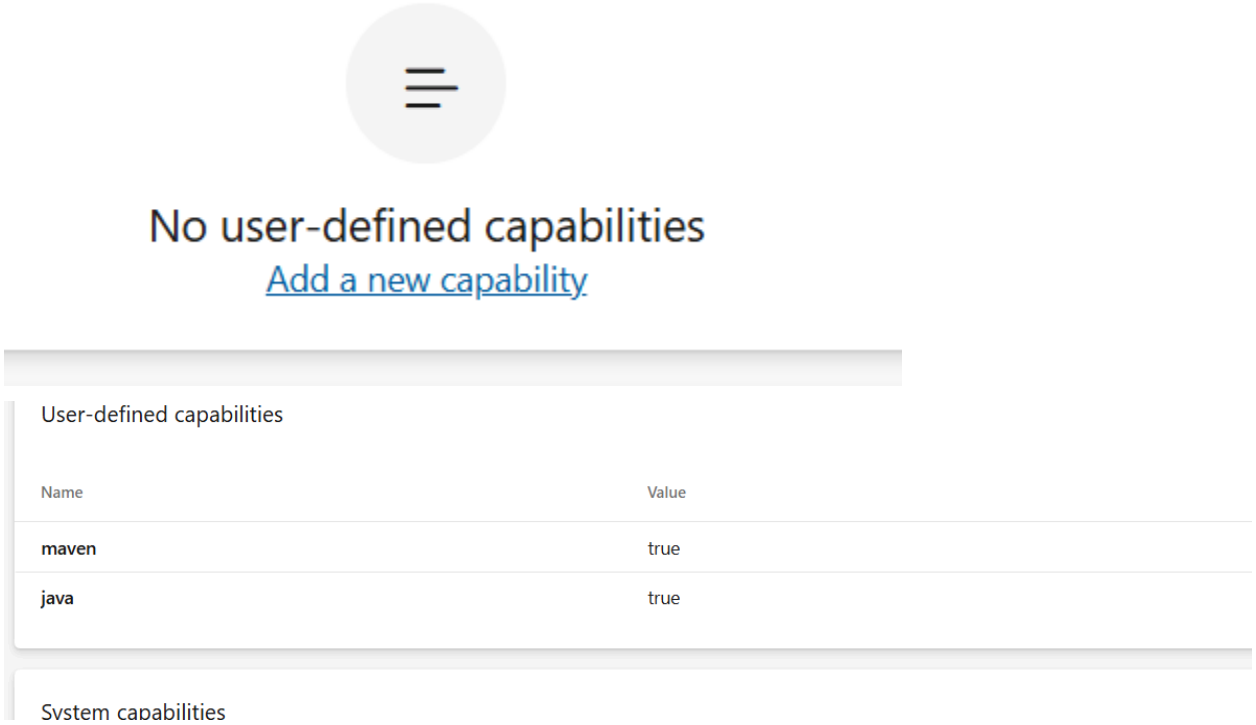
```
docker run -d --name sonar -p 9000:9000 mc1arke/sonarqube-with-community-branch-plugin
```

Our sonar qube is running >>username and pass is admin



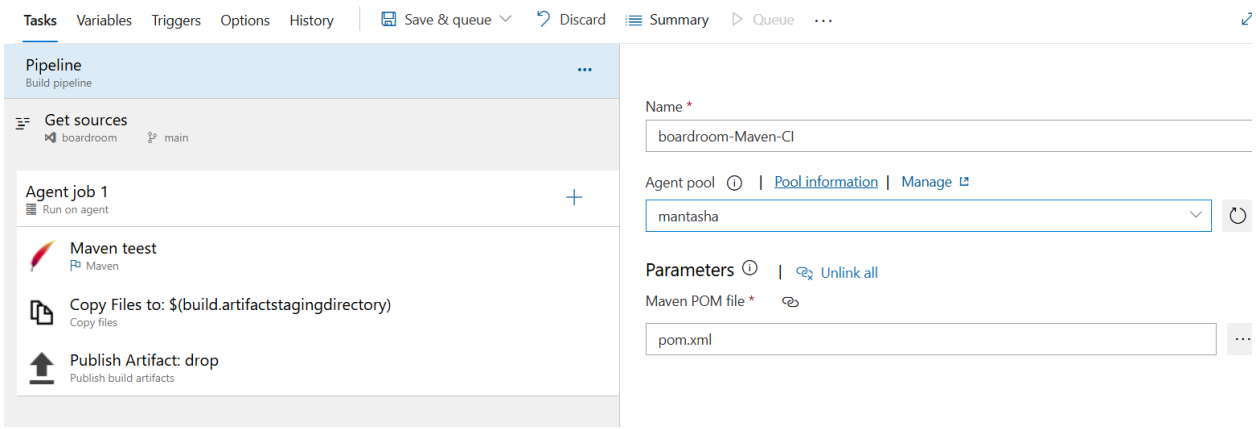
Run these commands

We need to add java and maven
So go to agent >> inside it >> capabilities

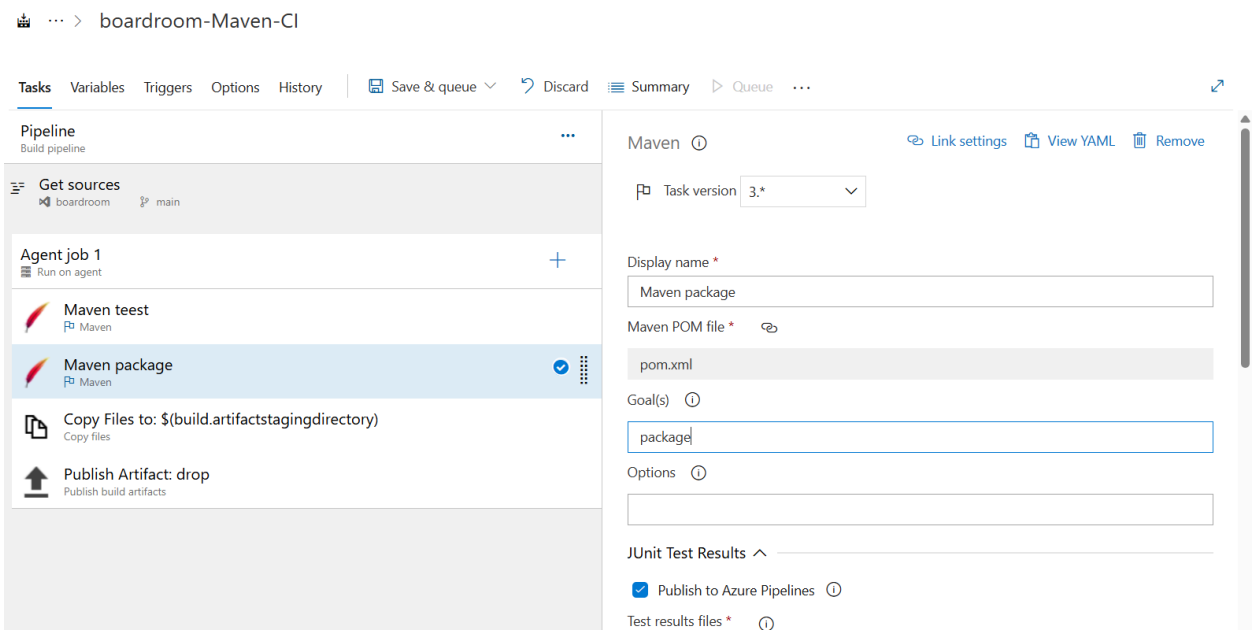


Now lets start the agents
./run.sh

Now go to your project >pipeline>edit

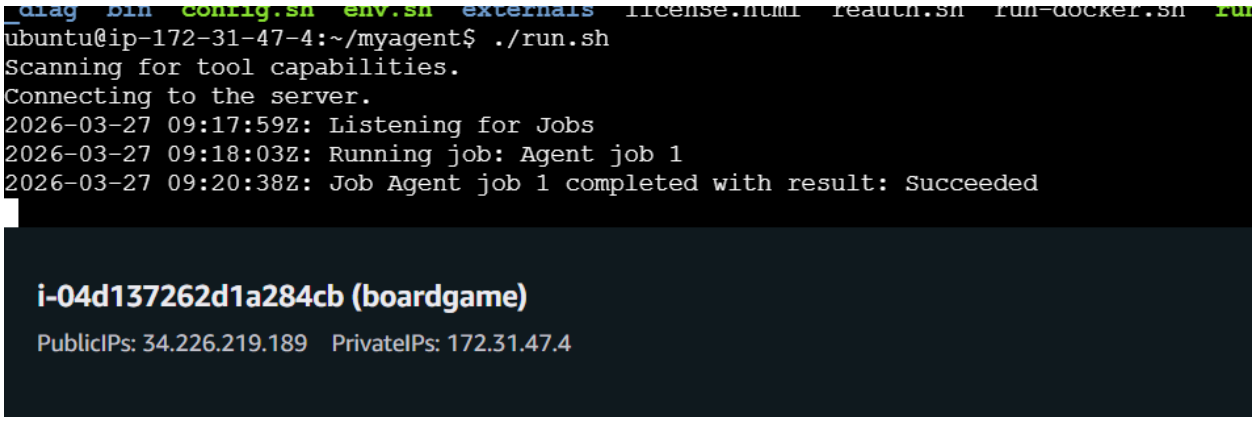


q

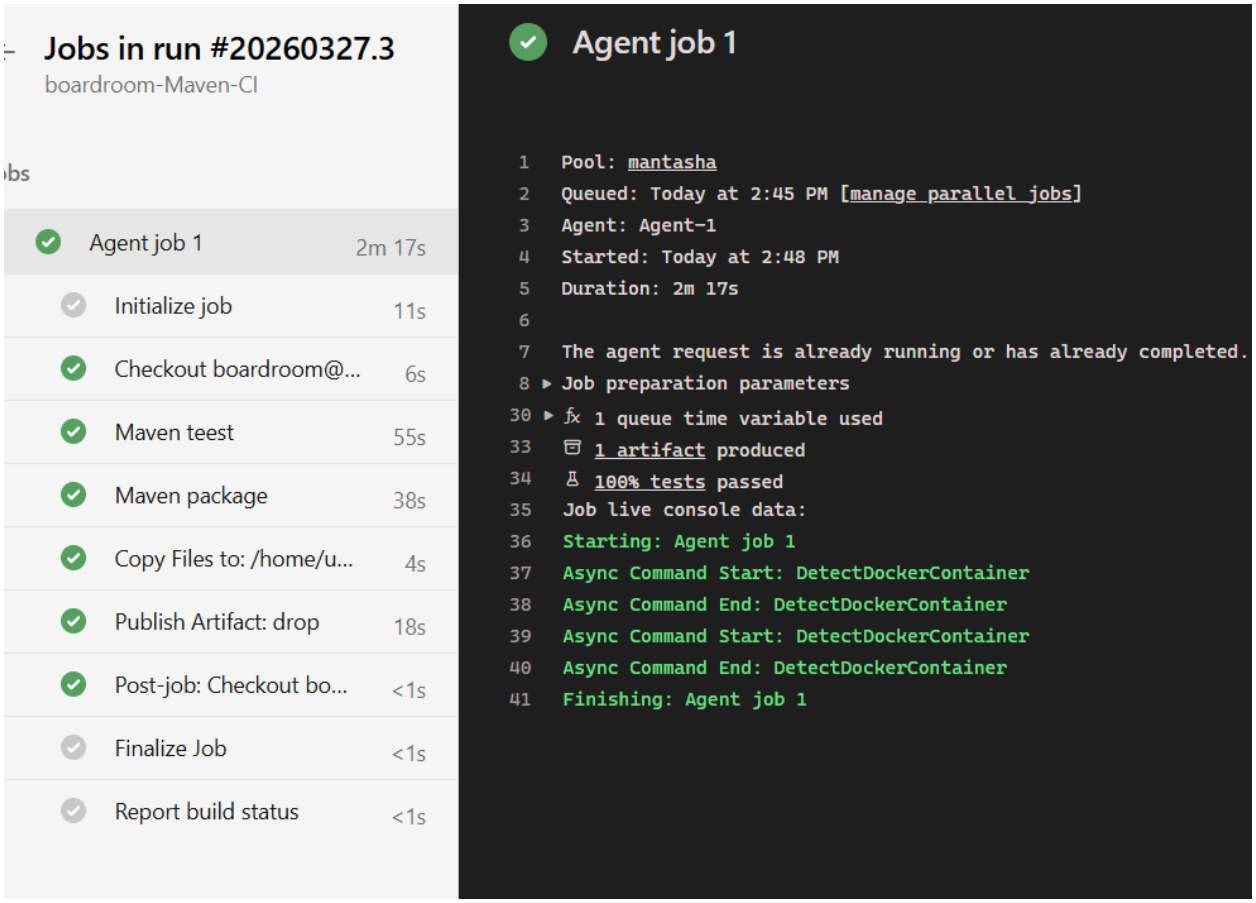


Now save and run

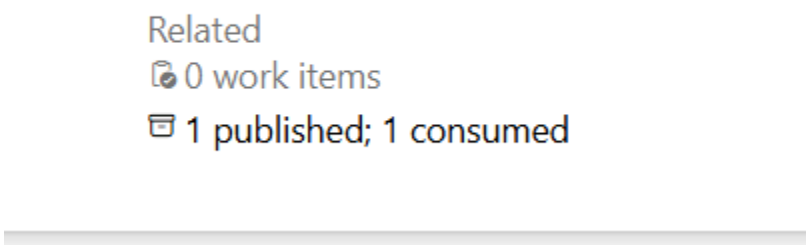
Now your pipeline will run



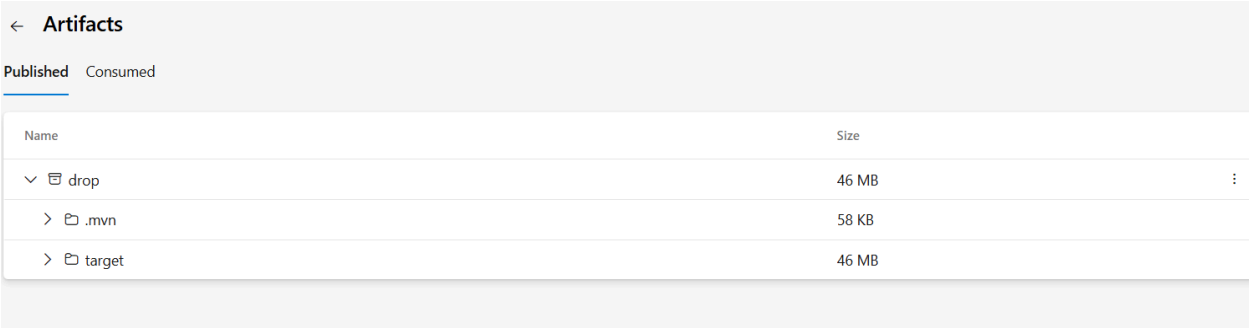
On your azure you can see its running



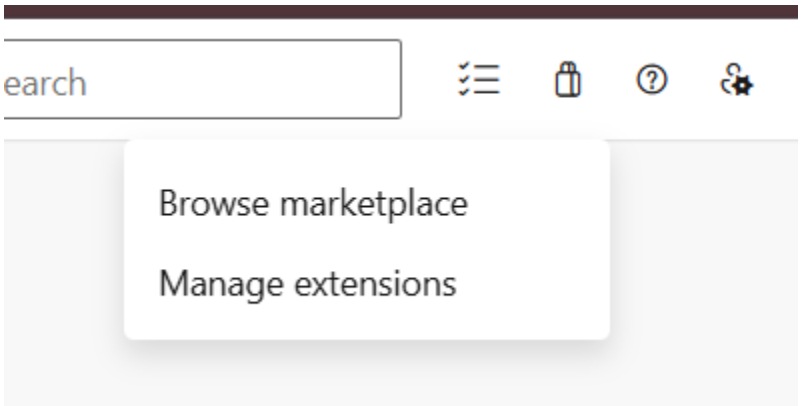
And its done go to artifac and you can see the articat here

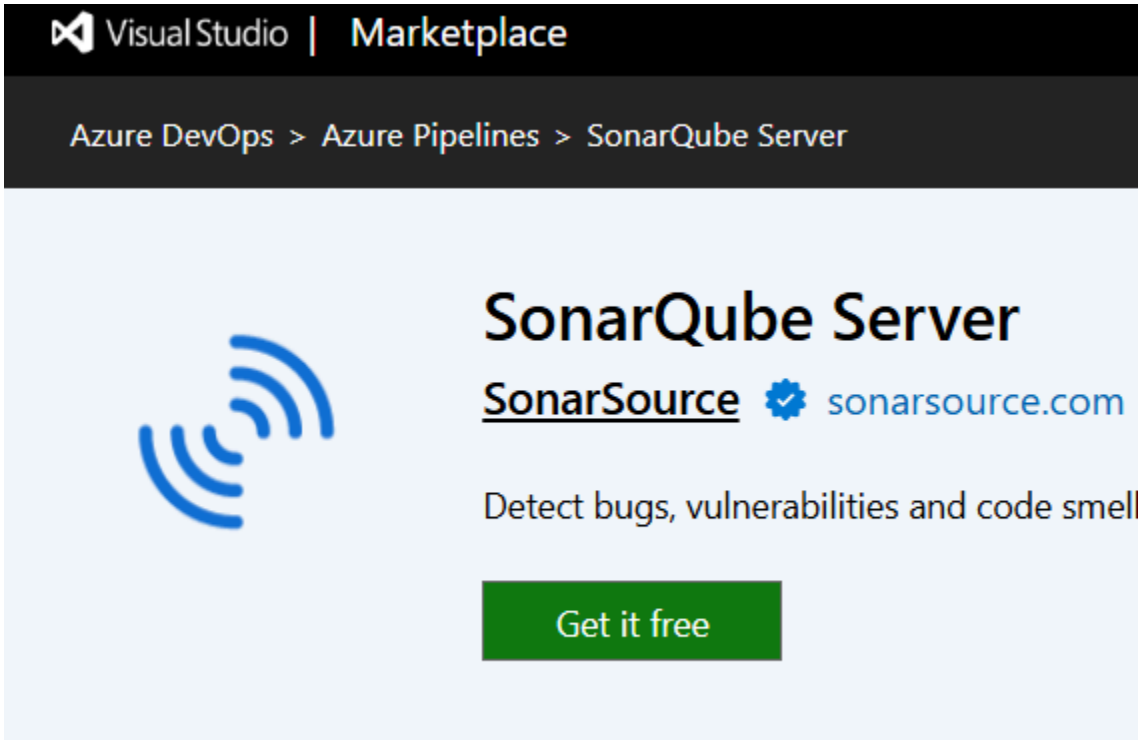


You can see the artifaces



Now lets integrate the sonar qube
Go to pipeline >> edit





Last updated

Mar 16, 2026 at 10:04 PM GMT+5:30

Permission scope

Build (read)
Code (read)
Test management (read)

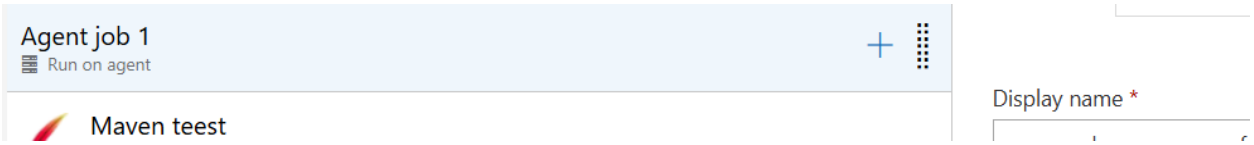
Select an Azure DevOps organization

azure0pipeline

Install

Aws is good for info
And for pipeline azure is better

Now let edit your pipeline >+ >>



Squ_1bc3c495f340cc7237721e1ee07bbfb60b7b99b3

New service connection

Server Url

http://34.226.219.189:9000/

Url for the SonarQube Server to connect to.

Authentication

Token

••

Authentication Token generated through SonarQube Server (go to My Account > Security > Generate Tokens)

Service connection details

Service Connection Name

son-cin

Description (optional)

Security

☐ Grant access permission to all pipelines

Learn more

Troubleshoot

Save

Save it

Prepare Analysis Configuration

Link settings | View YAML | Remove

Task version 8.*

Display name *

sonarqube prepare conf

SonarQube Server Endpoint * | Manage

son-cin

New

Choose the way to run the analysis *

- ○ Integrate with .NET
- ◉ Use standalone SonarScanner CLI
- ○ Integrate with Maven or Gradle

Project Key *

Boardgame

Project Name

Boardgame

Project Version

1.0

Sources directory root *

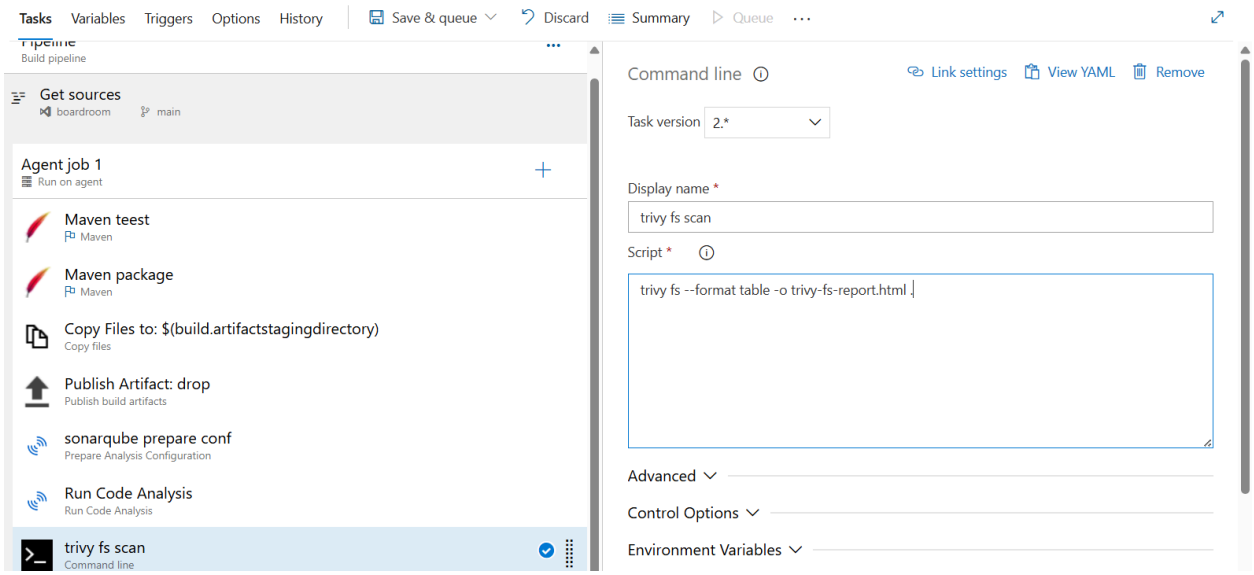
.

Advanced

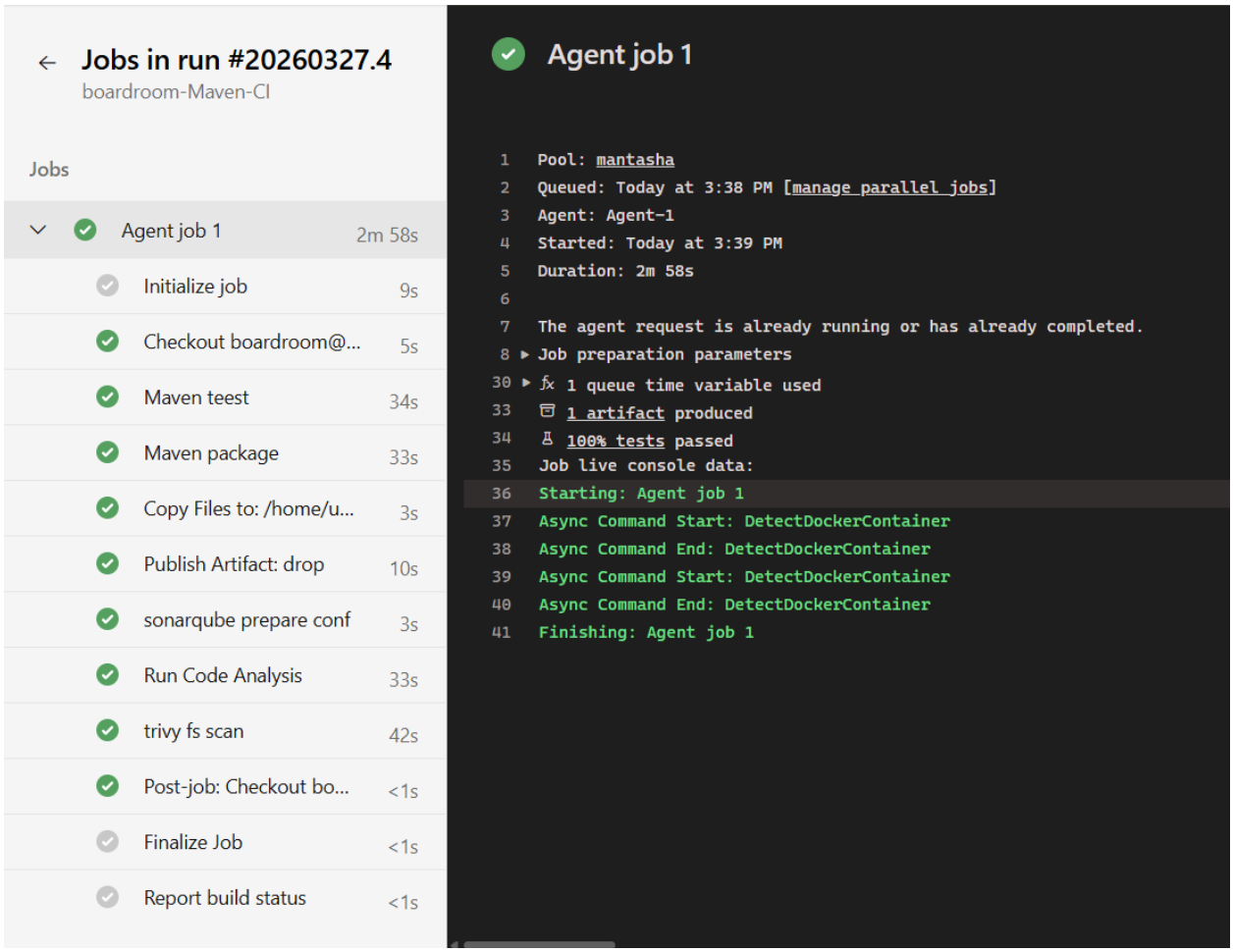
Additional Properties

```
# Additional properties that will be passed to the scanner,
# Put one key=value per line, example:
# sonar.exclusions=**/*.bin
sonar.java.binaries=target/
```

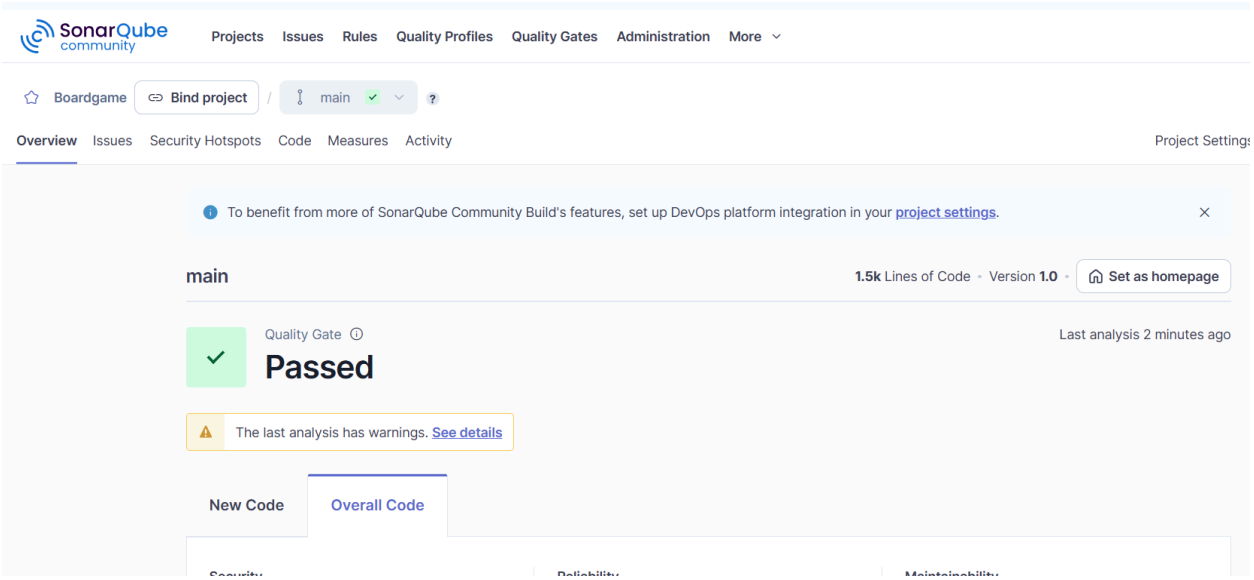
Now add one more
trivy fs --format table -o trivy-fs-report.html



Now save and run these and its done success fully



On our sonar qube we can see the output



Now we are going to add docker

New service connection

Registry type

Docker Hub Others Azure Container Registry

Docker Registry

https://index.docker.io/v1/

Docker ID

allenaira

Docker Password

••••••••••••••••••••••••••••••••••

Email (optional)

Verify Verification Succeeded

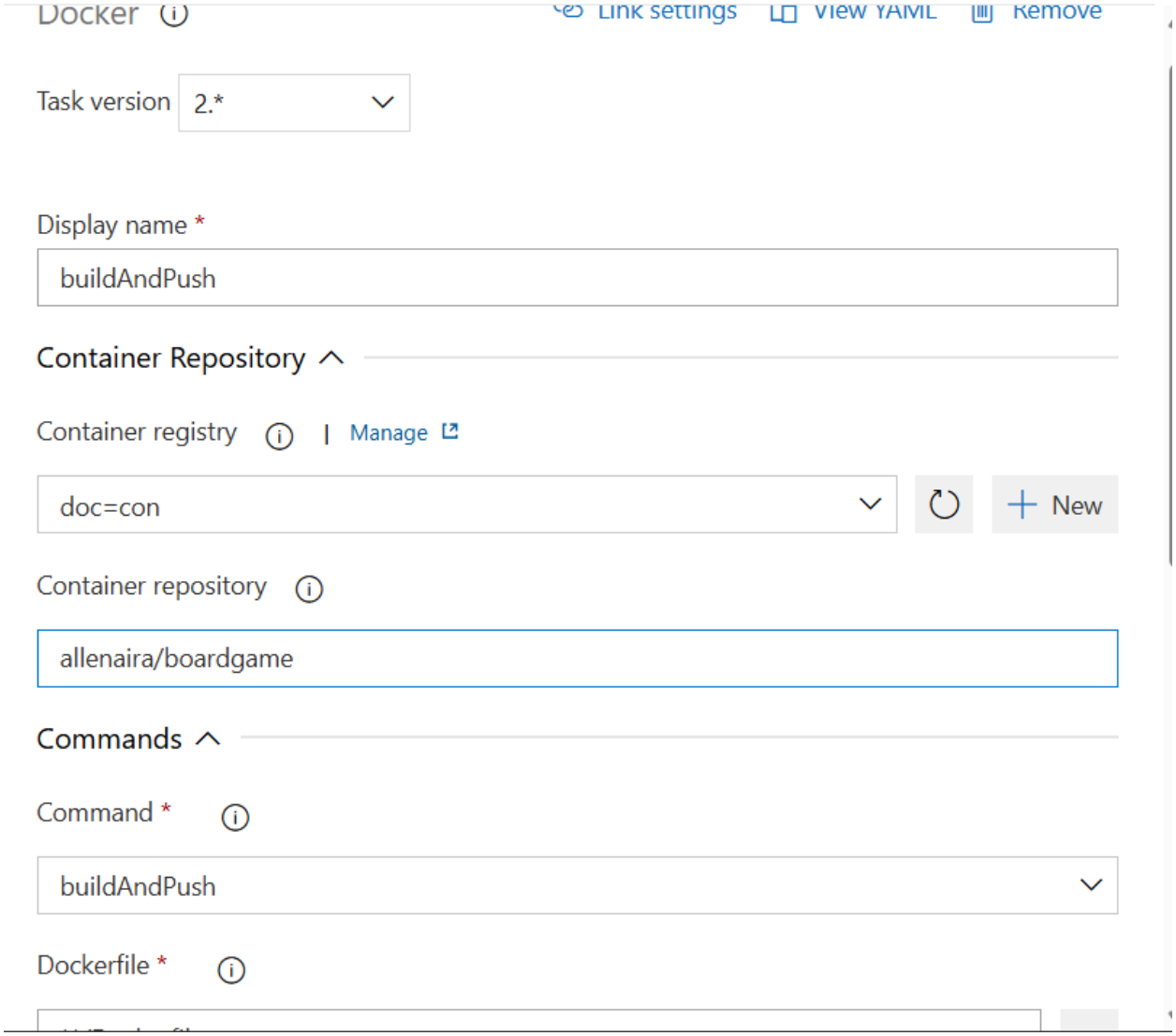
Service connection details

Service Connection Name

doc=con

Description (optional)

Add these connection



Save and refresh it and re sun it again

Make sure your ec2 server is running in background with [run.sh](run.sh) file

trivy image --format table -o trivy-image-report.html . allenaira/boardgame:latest

boardroom-Maven-CI

Tasks | Variables | Triggers | Options | History | Save & queue | Discard | Summary | Queue

Agent job 1
Run on agent

- Maven teest — Maven
- Maven package — Maven
- Copy Files to: $(build.artifactstagingdirectory) — Copy files
- Publish Artifact: drop — Publish build artifacts
- sonarqube prepare conf — Prepare Analysis Configuration
- Run Code Analysis — Run Code Analysis
- trivy fs scan — Command line
- buildAndPush — Docker
- trivy image scan docker

Task version 2.*

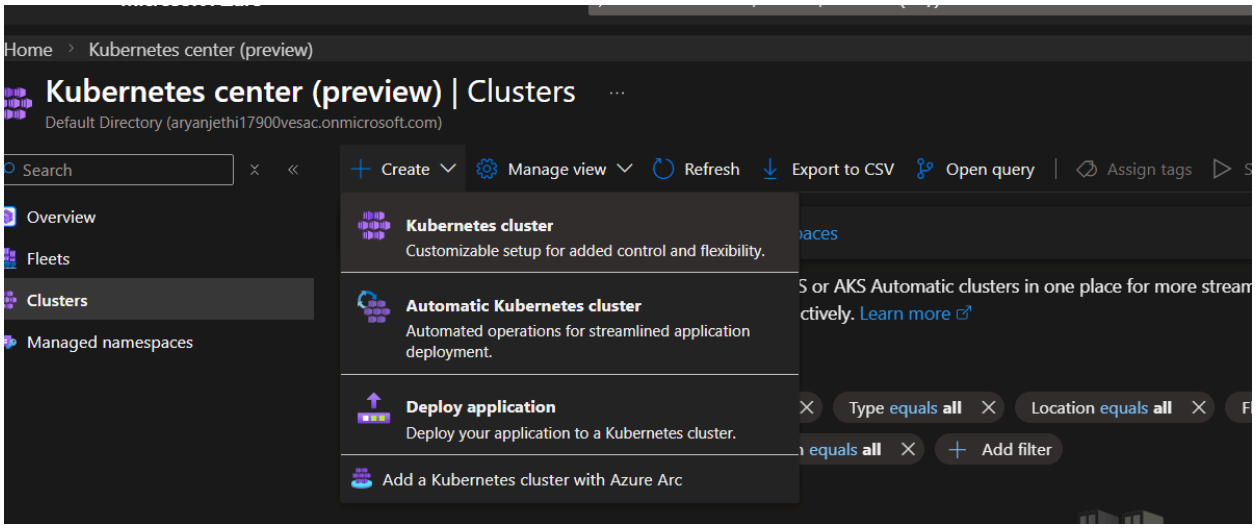
Display name *
trivy image scan docker

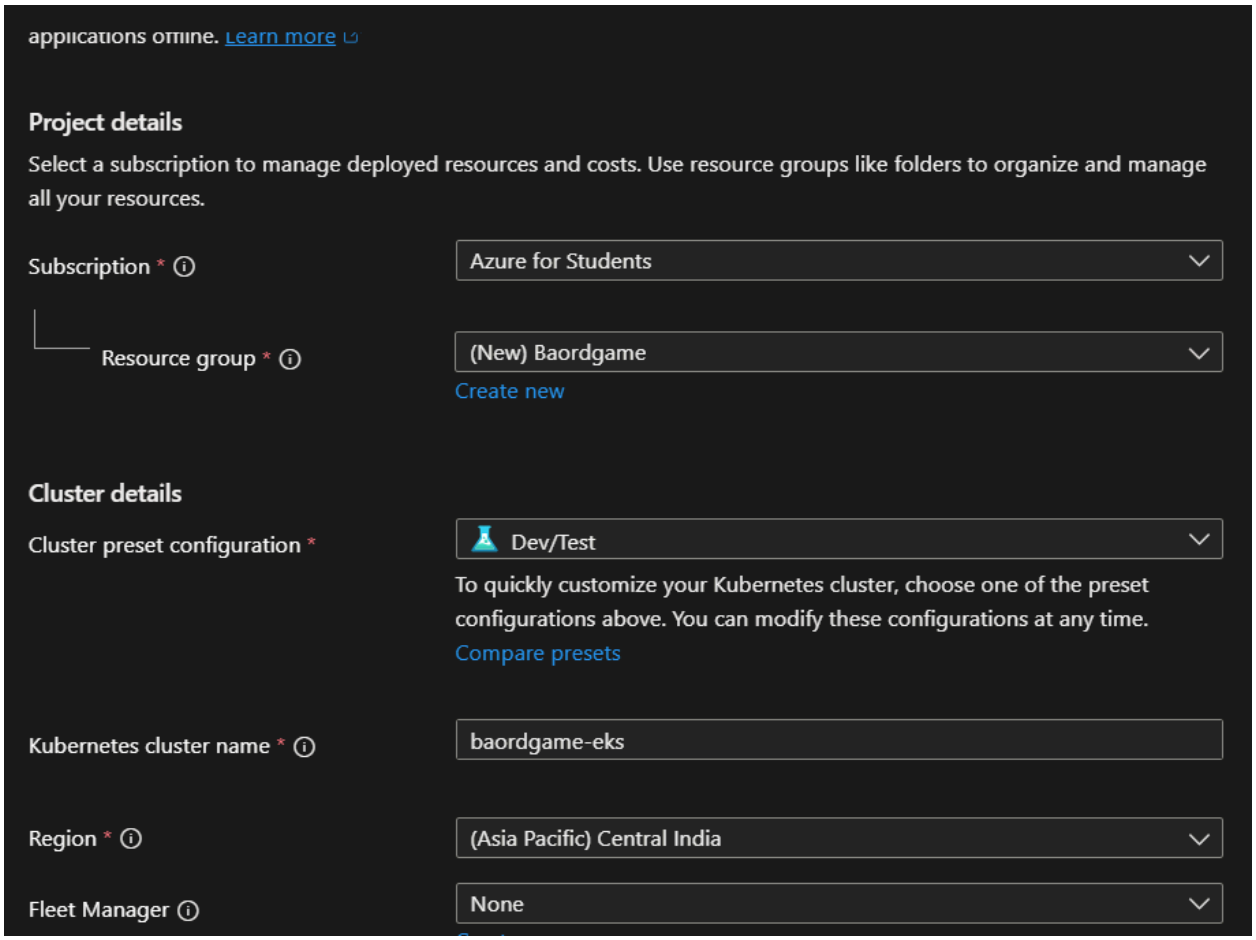
Script *
```
trivy image --format table -o trivy-image-report.html . allenaira/boardgame:latest
```

Advanced
Control Options
Environment Variables
Output Variables

```
Pool: mantasha
Queued: Today at 3:51 PM [manage parallel jobs]
Agent: Agent-1
Started: Today at 3:52 PM
Duration: 2m 48s

The agent request is already running or has already completed.
▶ Job preparation parameters
▶ fx 1 queue time variable used
1 artifact produced
100% tests passed
Job live console data:
Starting: Agent job 1
Async Command Start: DetectDockerContainer
Async Command End: DetectDockerContainer
Async Command Start: DetectDockerContainer
Async Command End: DetectDockerContainer
Finishing: Agent job 1
```

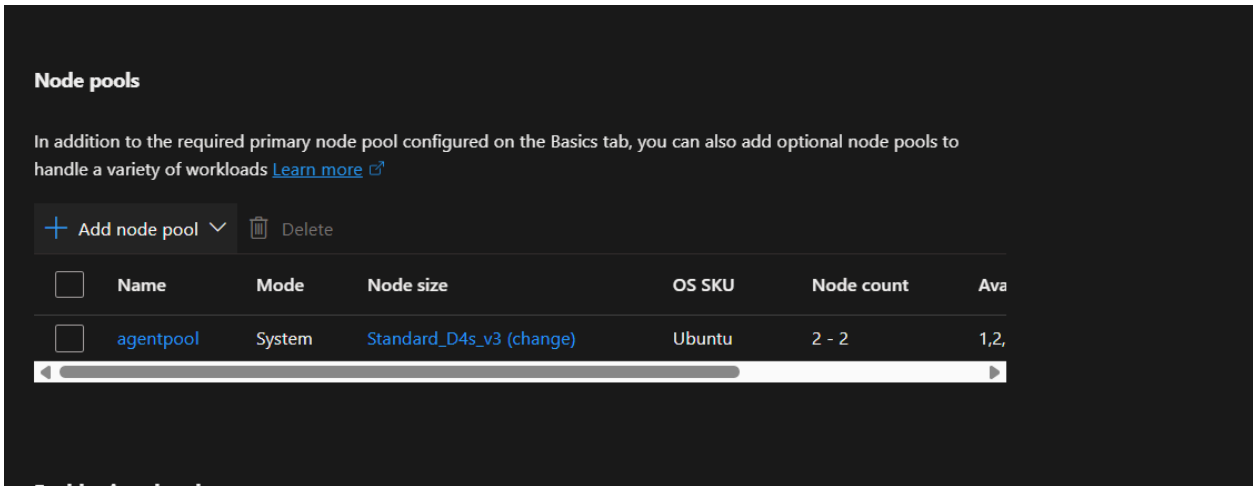
Now we are going to deploy it using AKS



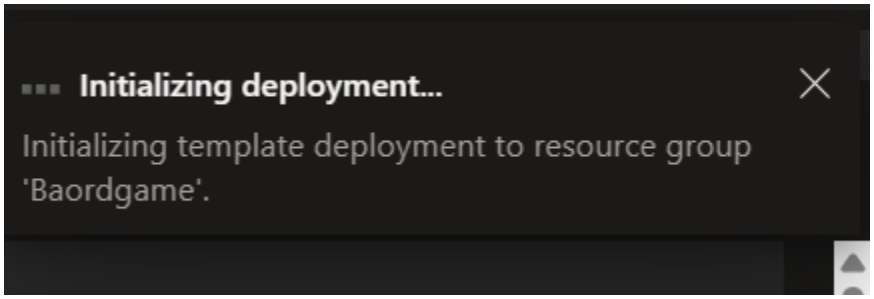


Here we need to do these
2 node pool size
Standar_d4s_v3 in case not work change to Standard_D2s_v3 (change) these

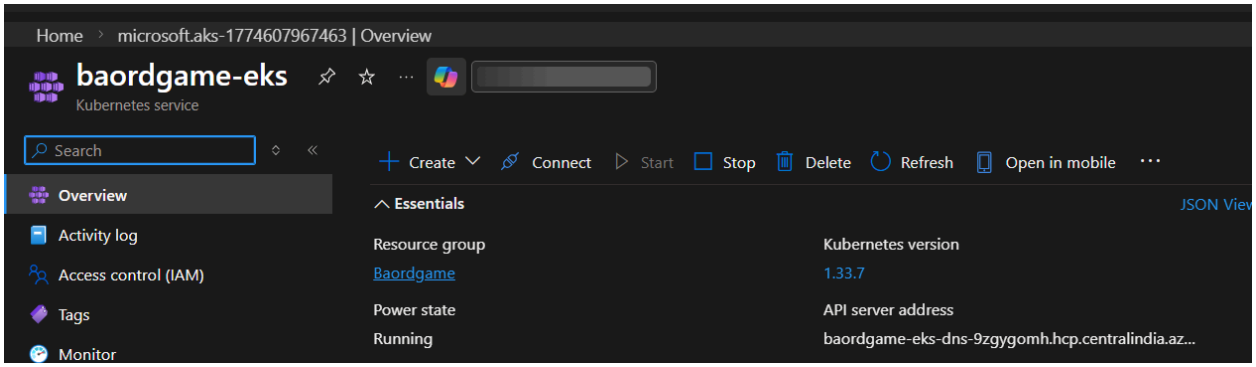
Or go with one or one node only
Change these



Review and create
And click on create



Your cluster is ready to use



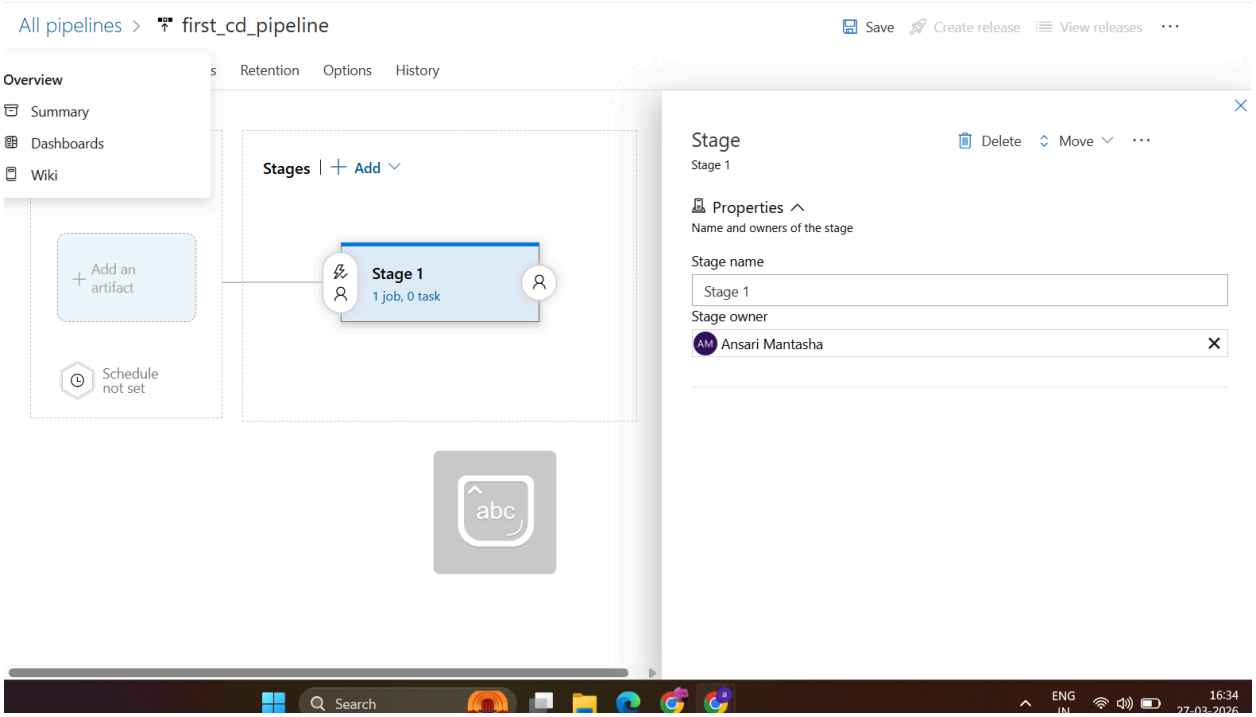
Everytime you run the project it will push the image to your docker hub

Once its done then go to

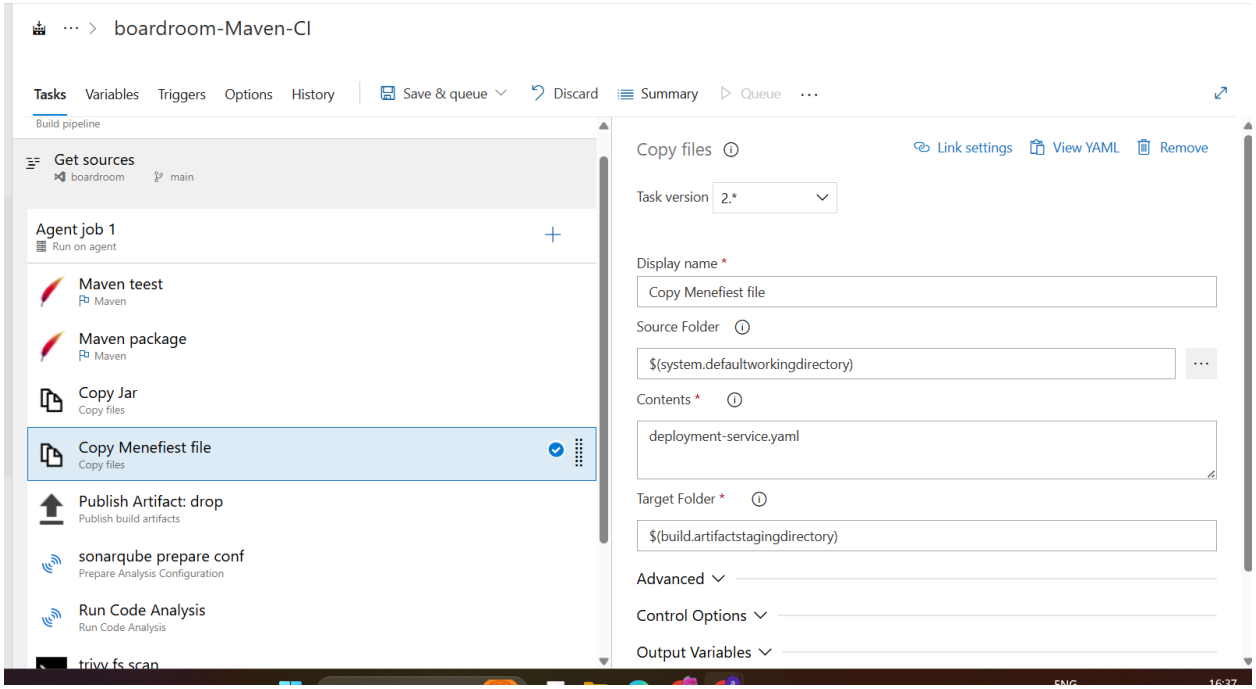
Now lets create a release pipeline

Go with empty job

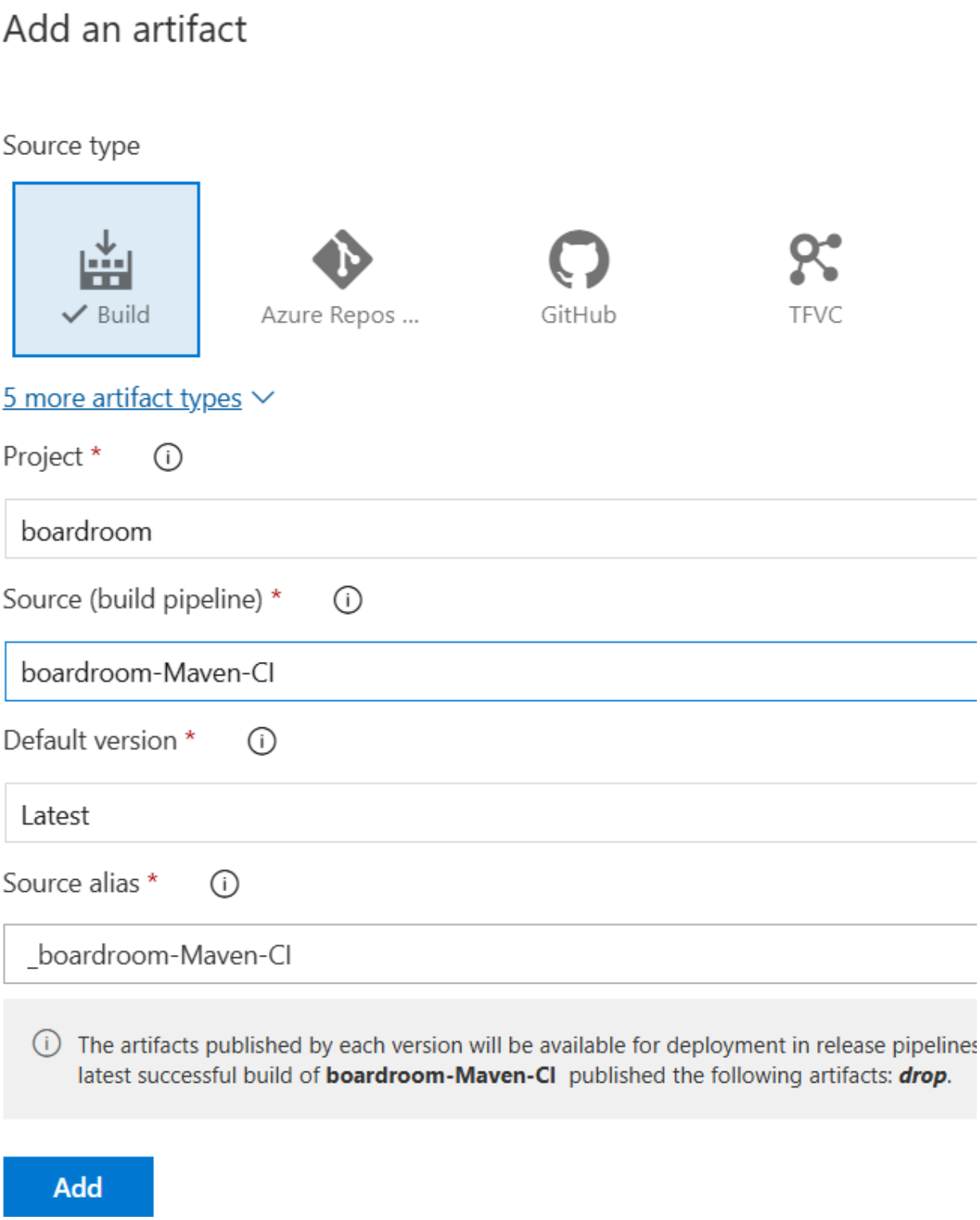
Here add an artiface



We need an artiface for that clone the artiface in your pipelinne



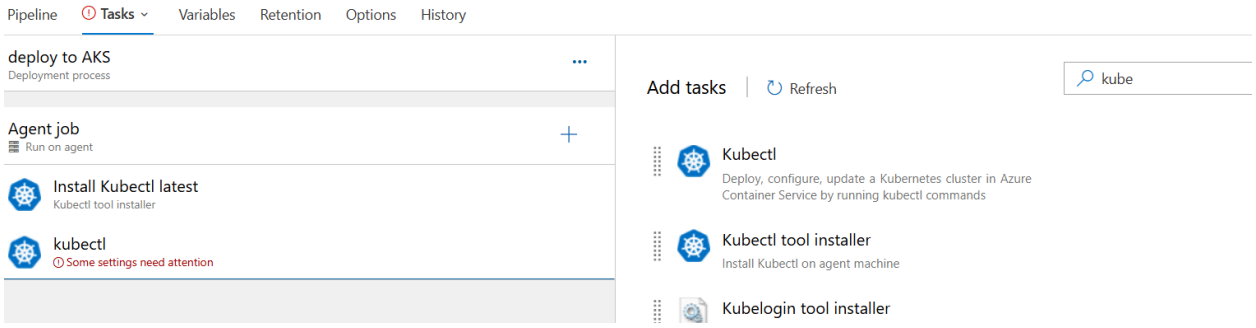
Add these very imp



Add these
After these click in thunder sign and addthese

Now click on stage

Name it >> agent
We didnot install kubectl so we can add it



Now lets create connections

New service connection

Authentication method

- ○ KubeConfig
- ○ Service Account
- ◉ Azure Subscription

Azure Subscription

Azure for Students (3dd77353-9376-4da6-b87d-cd1ae2f8b5...

Cluster

baordgame-eks (Baordgame)

Namespace

default

☑ Use cluster admin credentials

Service connection details

Service Connection Name

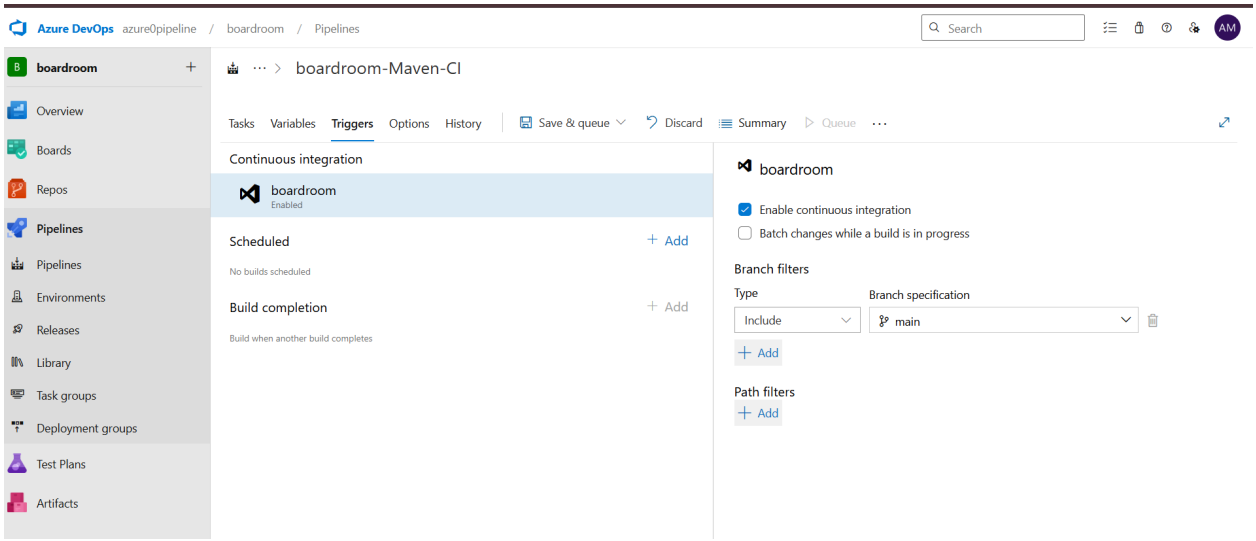
con-name

Description (optional)

Security

☑ Grant access permission to all pipelines

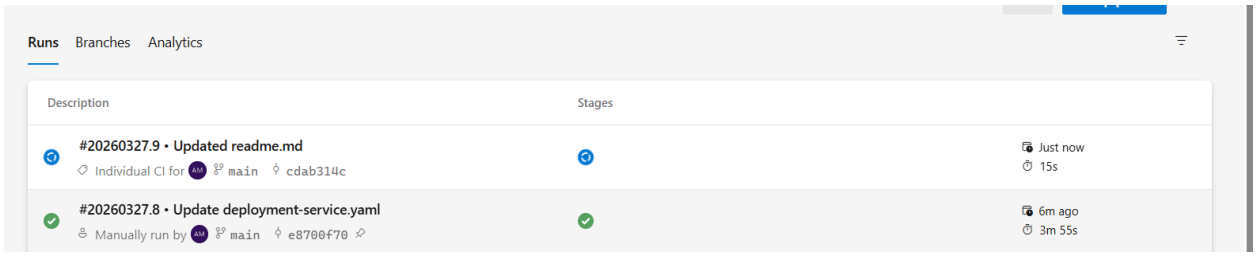
And save it



D othese aw well

Now run your pipeline so we can get the ym file og the the yml file hre

Now if we do anychange in our code our pipeline start triggering



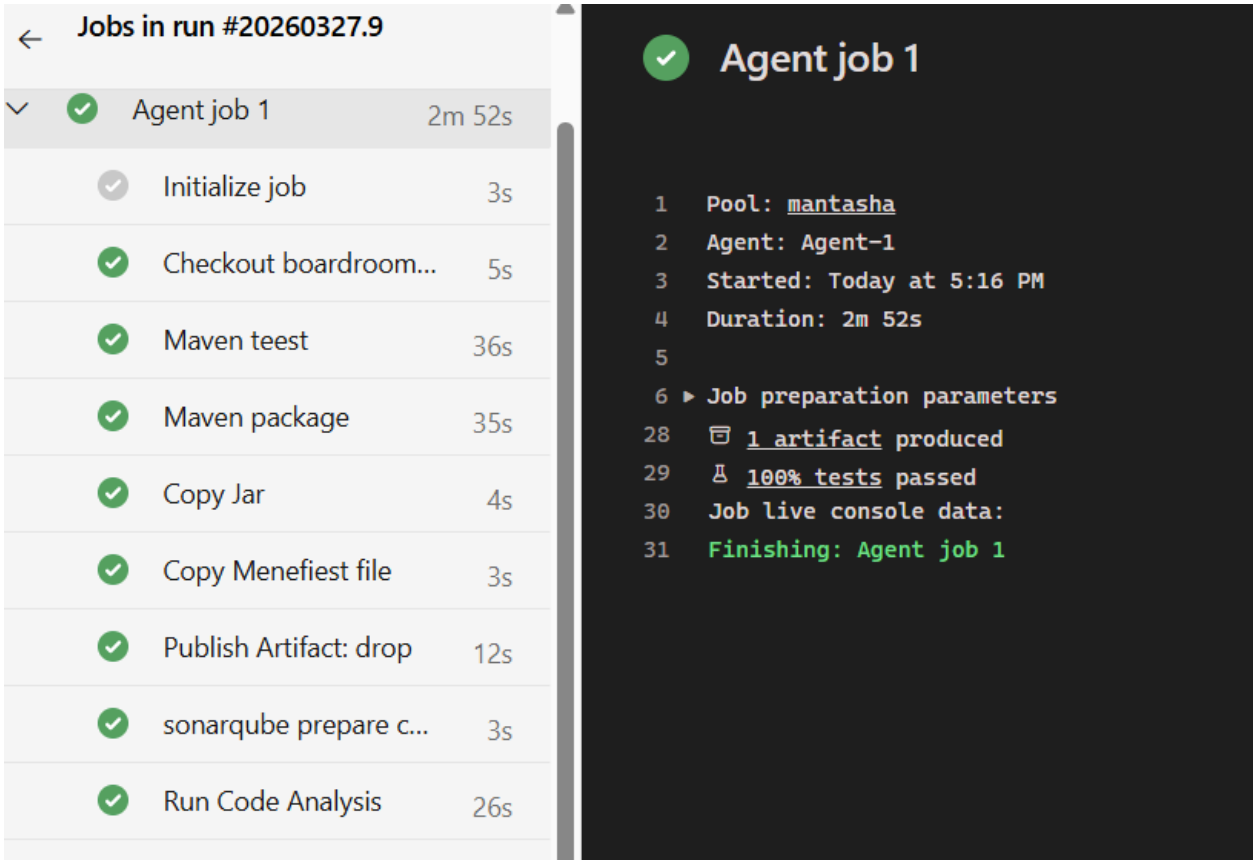
See

And

And on your server you cna see everything is up and runny

```
Last login: Fri Mar 27 11:33:24 2026 from 18.206.107.27
ubuntu@ip-172-31-47-4:~$ cd myagent/
ubuntu@ip-172-31-47-4:~/myagent$ ls
_diag  _work  bin  config.sh  env.sh  externals  license.html  reauth.sh  run-docker.sh  run.sh  svc.sh  vsts-agent-linux-x64-4.270.0.tar.gz
ubuntu@ip-172-31-47-4:~/myagent$ cd _work/
ubuntu@ip-172-31-47-4:~/myagent/_work$ ls
1  SourceRootMapping  _tasks  _temp  _tool
ubuntu@ip-172-31-47-4:~/myagent/_work$ cd 1
ubuntu@ip-172-31-47-4:~/myagent/_work/1$ ls
TestResults  a  b  s
ubuntu@ip-172-31-47-4:~/myagent/_work/1$ cd s
ubuntu@ip-172-31-47-4:~/myagent/_work/1/s$ ls
Dockerfile   deployment-service.yaml  mvnw.cmd  readme.md                 src     trivy-fs-report.html
Jenkinsfile  mvnw                     pom.xml   sonar-project.properties  target  trivy-image-report.html
ubuntu@ip-172-31-47-4:~/myagent/_work/1/s$
```

Once your ci run then your cd will run



And your release s running now

board_cd > Release-1 > Stage 2 In progress

Pipeline Tasks Variables Logs Tests Deploy Cancel Refresh Download all logs Edit ···

Deployment attempt #2
progress

Agent job
In progress

Agent job

Pool: mantasha · Agent: Agent-1

Initialize job

```
********************************************************************************
Starting: Initialize job
********************************************************************************
Agent name: 'Agent-1'
Agent machine name: 'ip-172-31-47-4'
Current agent version: '4.270.0'
Agent running as: 'ubuntu'
Prepare release directory.
ReleaseId=1, TeamProjectId=4d0bc579-34e6-4264-af01-693ed28262a3, ReleaseDefinitionName=boar
Release folder: /home/ubuntu/myagent/_work/r1/a
Environment variables available are below.  Note that these environment variables can be re
e ReleaseDefinition) by replacing "_" with "." e.g. AGENT_NAME environment variable can be
in the ReleaseDefinition:
                                [AGENT_ACCEPTTEEEULA] --> [True]
                                [AGENT_CLOUDID] --> []
                                [AGENT_DISABLELOGPLUGIN_TESTFILEPUBLISHERPLUGIN] --> [true]
                                [AGENT_DISABLELOGPLUGIN_TESTRESULTLOGPLUGIN] --> [true]
                                [AGENT_HOMEDIRECTORY] --> [/home/ubuntu/myagent]
                                [AGENT_ID] --> [9]
                                [AGENT_ISSELFHOSTED] --> [1]
                                [AGENT_JOBNAME] --> [Agent job]
                                [AGENT_MACHINENAME] --> [ip-172-31-47-4]
                                [AGENT_NAME] --> [Agent-1]
                                [AGENT_OS] --> [Linux]
```

In your kubernetes you can see your pods up and running

Home > baordgame-eks

baordgame-eks | Workloads

Kubernetes service

Search

Overview
Activity log
Access control (IAM)
Tags
Monitor
Diagnose and solve problems
Microsoft Defender for Cloud
Cost analysis
Resource visualizer
Kubernetes resources
Namespaces
Workloads
Services and ingresses
Network policies
Storage
Configuration
Custom resources

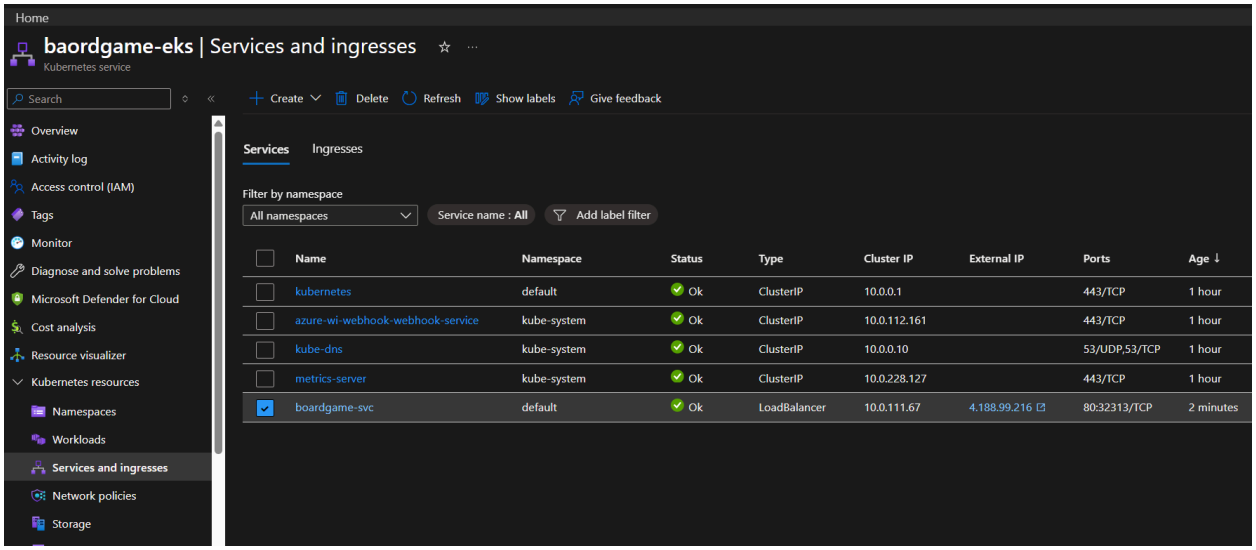
Create Scale Delete Refresh Show labels Give feedback

Deployments Pods Replica sets Stateful sets Daemon sets Jobs Cron jobs

Filter by namespace: All namespaces | Deployment name : All | Add label filter

Name	Namespace	Ready	Age	CPU	Memory
azure-wi-webhook-controller-manager	kube-system	2/2	1 hour	Enable metrics	
eraser-controller-manager	kube-system	1/1	1 hour	%	%
coredns	kube-system	2/2	1 hour	%	%
coredns-autoscaler	kube-system	1/1	1 hour	%	%
konnectivity-agent	kube-system	2/2	1 hour	%	%
konnectivity-agent-autoscaler	kube-system	1/1	1 hour	%	%
metrics-server	kube-system	2/2	1 hour	%	%
boardgame-deployment	default	2/2	32 seconds	%	%

To access your website you need to click on the link given here



And boom you app is up and running

Now lets delete everything

Its is a end to end pipeline where

Kubernetes Service Connection

Kubernetes service connection * | Manage

con-name

New

Namespace

default

Commands

Command

apply

Use configuration

Configuration type

File path

Inline configuration

File path *

$(System.DefaultWorkingDirectory)/_boardroom-Maven-CI/drop/deployment-service.yaml

Arguments